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Bulk-Surface Models for Ligand-Receptor Interactions

Bachelor Thesis in Mathematics

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Abstract

Interactions between ligands and receptors can be influenced by the heterogeneous distribution of receptors on the cell membrane. We study several theoretical models describing these ligand-receptor dynamics both analytically, by showing the existence and uniqueness of non-negative and bounded weak solutions for finite time intervals, and numerically, by discussing an implementation of the finite element method using FEniCSx. These numerical simulations show that it is possible to use theory on Turing patterns to explain the spatial effects in distribution of receptors on the cell membrane. We derive a reduced variant of the model that is both simpler to study analytically and less computationally expensive to simulate, and show that this reduced model is an adequate representation of its full counterpart.

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Lastly, as I write this thesis to end my time at Radboud University, I want to thank my family, my old friends from Wageningen, as well as my new friends whom I have met during my time in Nijmegen, for supporting me in moments of stress, for listening to me ramble about dots and stripes appearing across some digital spheres, and for providing a seemingly endless supply of brownies during the late nights at Huize Duke.

Use of AI

Large language models were used for grammar and spell-checking in this thesis.

Data management

All simulations in this thesis were written in Python using the FEniCSx library. The results were either directly plotted with PyVista or written to XDMF files to be plotted with ParaView. The scripts for both mesh generation with Gmsh and simulation with FEniCSx are available on the public GitHub page `https://github.com/bregtvdl03/Bachelor_Thesis_Turing_patterns`. Due to their size, the complete smiulation output files are not included on this page, but they are stored locally on the author's personal computer.

Contents

Introduction

Many biological processes can be accurately described by partial differential equations (PDEs). In particular, reaction-diffusion equations can be used to model the interactions in a system of chemical substances spreading through a domain and reacting with each other. In this thesis, we study various models for a biological process related to cell communication.

Cells can communicate with one another by producing signalling molecules referred to as *ligands* and releasing them into their environment. These ligands travel through the extracellular space until they are captured by a *receptor* on the membrane of another cell, forming a bounded receptor, or ligand-receptor *complex*. Specifically, we consider the case where these receptors couple to a *G-protein*. These receptors can diffuse across the cell membrane. When a ligand binds to a receptor, a reaction is triggered that passes the signal through the membrane and activates a G-protein, which further passes the signal to the interior of the cell, where it can be processed appropriately. Eventually, the ligand is released by the receptor, enabling the ligand and receptor to repeat the process of diffusing across their domains and reacting with other receptors and ligands, respectively.

Due to their diffusing nature, one might expect to find G-protein coupled receptors homogeneously distributed across the cell membrane. However, experiments have shown that this is not the case. Instead, they form clusters of various sizes, which can influence their signalling behaviour [KLS+24, MFM25]. We aim to formulate and study reaction-diffusion systems predicting such clustering using bulk-surface models, since receptors are confined to the cell membranes.

In this thesis, we study various systems of bulk-surface systems modelling the concentrations of ligands, receptors and their complex. Chapter 1 establishes the required mathematical background in the field of PDEs, introducing notions such as weak derivatives and weak solutions to PDEs, as well as recalling some simple but important inequalities that are essential in later proofs. Chapter 2 derives a model involving ordinary differential equations (ODEs) for the ligand-receptor interactions with assumptions from [KMI14] and discusses a classical proof of existence and uniqueness of weak solutions to his model, as well as some qualitative properties. Chapter 3 introduces the theory of Turing patterns, which is often used to account for spatial organisation in mathematical biology. Chapter 4 introduces the finite element method to simulate another variant of model. Using these simulations, we seek to explain the formation of receptor clusters by this model's ability to generate Turing patterns. We also reduce the model to a system of fewer components and analyse how well this reduced variant represents the full system.

This thesis is written under the assumption that the reader is comfortable with mathematical theory typically taught in a mathematics bachelor's programme. In particular, relative fluency in subjects such as multivariate calculus, linear algebra and ODEs is required. Knowledge on the fundamental definitions and theory of PDEs, in particular the diffusion equation, and functional analysis is useful.

Chapter 1

Preliminaries

1.1 Analytical background and notation

We begin by specifying some notation and recalling and defining some general tools that play an important role throughout this thesis. We will be working with a wide variety of spaces, many of them having their own inner products and norms. First of all, we denote the Euclidean norm on $\mathbb{R}^m$ by $|\cdot|$, for all $m \in \mathbb{N}$. For a more general normed space X, where X is different from $\mathbb{R}^m$ for some m, we write $\|\cdot\|_X$ for its norm. When this norm is induced by an inner product, we denote this inner product by $(\cdot\,,\cdot)_X$. First we collect a few inequalities that are mainly used in the estimates of our existence and uniqueness proof.

Lemma 1.1 (**Young's inequality**)
For every $a, b \in \mathbb{R}$, we have

$$ab \leq \frac{1}{2}\left(a^2 + b^2\right).$$

More generally, for every $\varepsilon > 0$, we have

$$ab \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2.$$

Proof. Let $a, b \in \mathbb{R}$ and $\varepsilon > 0$. Then

$$0 \leq \left(\sqrt{\varepsilon} a - \frac{1}{2\sqrt{\varepsilon}} b\right)^2 = \varepsilon a^2 - ab + \frac{1}{4\varepsilon} b^2.$$

This proves the second inequality. The first inequality is a special case of the second, with ε set to $\frac{1}{2}$. □

To formulate the next inequality in its most general form, we need a new notion of continuity.

Definition 1.2
Let $a < b$. A function $f : [a, b] \to \mathbb{R}$ is called *absolutely continuous* if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of subintervals $\{(x_k, y_k)\}_{k=1}^N$ of $[a, b]$,

$$\sum_{k=1}^{N}(y_k - x_k) < \delta \quad \implies \quad \sum_{k=1}^{N}|f(y_k) - f(x_k)| < \varepsilon.$$

This notion of continuity is much stronger than the traditional one. There is a crucial equivalence, proved in [Nie97].

Proposition 1.3
A function $f : [a, b] \to \mathbb{R}$ is absolutely continuous if and only if f has derivative f' almost everywhere, f' is integrable, and

$$f(x) = f(a) + \int_a^x f'(t)\,\mathrm{d}t$$

for every $x \in [a, b]$.

We can now formulate Gronwall's lemma, which is used very often in the study of differential equations to derive explicit estimates given an inequality that involves the unknown function.

Lemma 1.4 (Gronwall's lemma: differential form)
Let $T > 0$ and let η be a non-negative, absolutely continuous function on $[0, T]$, which satisfies

$$\eta'(t) \leq \varphi(t)\eta(t) + \psi(t)$$

for almost every $t \in [0, T]$, where φ and ψ are non-negative, L^1 functions on $[0, T]$. Then

$$\eta(t) \leq e^{\int_0^t \varphi(s)\,\mathrm{d}s}\left[\eta(0) + \int_0^t \psi(s)\,\mathrm{d}s\right].$$

for all $t \in [0, T]$.

This is known as the *differential form* of Gronwall's lemma. A proof can be found in [Eva10]. We also have an *integral form*, which is especially useful when dealing with non-differentiable functions.

Lemma 1.5 (Gronwall's lemma: integral form)
Let $T > 0$ and let $y : [0, T] \to \mathbb{R}$ and $a : [0, T] \to \mathbb{R}$ be nonnegative functions such that a and ay belong to $L^1([0, T])$ and

$$y(t) \leq C + \int_0^t y(s)a(s)\,\mathrm{d}s$$

for some constant C and all $t \in [0, T]$. Then for any $t \in [0, T]$, we have

$$y(t) \leq Ce^{\int_0^t a(s)\,\mathrm{d}s}.$$

This formulation can be found in [CV02].

Definition 1.6
Let $\Omega \subseteq \mathbb{R}^d$ be open. A Lebesgue measurable function $f : \Omega \to \mathbb{R}$ is said to be *locally integrable* if

$$\int_K |f| < \infty$$

for all compact subsets K of Ω. The space of all locally integrable functions is denoted by $L^1_{\text{loc}}(\Omega)$, where we identify two functions with each other if they agree almost everywhere in Ω.

An important fact that should be noted is that every function that is square-integrable is also locally integrable. That is, $L^2(\Omega) \subseteq L^1_{\text{loc}}(\Omega)$. For the sake of completeness, we will sometimes state definitions or results with the hypothesis that a function is locally integrable, only to use them on functions that are square-integrable.

We now continue with some concepts that are often used to translate local properties into global results. This method plays a fundamental part in our later discussions about the formulation and analysis of weak forms and solutions to PDEs.

Let X be a topological space. A function $\varphi : X \to \mathbb{R}$ is said to have *compact support* if the set

$$\operatorname{supp}(\varphi) := \overline{\{x \in X \mid \varphi(x) \neq 0\}}$$

is a compact subset of X. For any open set $\Omega \subseteq \mathbb{R}^d$, the space of all infinitely-times continuously differentiable functions with compact support in Ω is denoted by $C_c^\infty(\Omega)$.

An important property of these functions is that they must be 0 on the boundary of Ω. In the particular context of PDE analysis, they are often referred to as *test functions*. They are used in a wide range of applications, for instance in the field of the Calculus of Variations. A fundamental tool in this theory is the following result, hence its name.

Lemma 1.7 (Fundamental Lemma of the Calculus of Variations)
Let $\Omega \subseteq \mathbb{R}^d$ be open and $f \in L^1_{\text{loc}}(\Omega)$. If

$$\int_\Omega f\varphi = 0$$

for all $\varphi \in C_c^\infty(\Omega)$, then $f = 0$ almost everywhere in Ω.

Proof. For every $\varepsilon > 0$, define $\eta_\varepsilon \in C_c^\infty$ as the standard mollifier from [Eva10, Appendix C.5]. Then Theorem 7 from [Eva10, Appendix C.5] implies that

$$\int_\Omega \eta_\varepsilon(x-y)f(y)\,\mathrm{d}y \to f(x) \qquad \text{as } \varepsilon \to 0$$

for almost every $x \in \Omega$. But since for small enough ε the function $y \mapsto \eta_\varepsilon(x-y)$ is in $C_c^\infty(\Omega)$, this limit is 0 due to the assumption, i.e. $f(x) = 0$ for almost every $x \in \Omega$. □

Multiplying a PDE with a test function and integrating by parts is sometimes referred to as *testing against* a test function $\varphi \in C_c^\infty(\Omega)$. We will see more applications of this in Section 1.3.

When we are differentiating functions in C^k defined in a subset of $\mathbb{R}^d$, we sometimes use the abbreviated notation

$$D^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}},$$

where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is called a *multi-index* and $|\alpha| := \alpha_1 + \dots + \alpha_d \leq k$.

1.2 Carathéodory theory

Next, we develop a weakened and more general notion of solutions to ODEs.

Definition 1.8
Let n be a positive integer, I an interval, $x : I \to \mathbb{R}^n$ and $f : S \to \mathbb{R}^n$ for some set $S \subseteq \mathbb{R} \times \mathbb{R}^n$. Then x is called a *solution of*

$$x'(t) = f(t, x(t)) \tag{1.1}$$

in the sense of Carathéodory on I if x is absolutely continuous, $(t, x(t)) \in S$ for every $t \in I$ and Equation (1.1) is satisfied for almost every $t \in I$.

Since this definition requires less from the solution, there is also a more general existence theorem, found in [CL55].

Theorem 1.9 (Carathéodory's existence theorem)
Let n be a positive integer, $a, b > 0$ and $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Define

$$R := \{(t, x) \in \mathbb{R} \times \mathbb{R}^n \mid |t - t_0| \leq a \text{ and } |x - x_0| \leq b\}$$

and let $f : R \to \mathbb{R}^n$ be continuous in x for every fixed t, measurable in t for every fixed x and bounded by some time-dependent integrable function m,

$$|f(t, x)| \leq m(t)$$

for all $(t, x) \in R$. Then the differential equation

$$x'(t) = f(t, x(t))$$

has a solution in the sense of Carathéodory on the interval $|t - t_0| \leq \beta$ for some $\beta > 0$, satisfying $x(0) = x_0$.

1.3 Weak derivatives and Sobolev spaces

Before we talk about weak solutions to PDEs, we need to define spaces of functions that can form potential candidates for solutions. We derive this space by weakening the conditions of differentiable functions.

Consider an open and bounded domain $\Omega \subseteq \mathbb{R}^d$. Then for any continuously differentiable function $u \in C^1(\Omega)$ and test function $\varphi \in C_c^\infty(\Omega)$, we have

$$\int_\Omega u_{x_i} \varphi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \varphi \, \mathrm{d}S - \int_\Omega u \varphi_{x_i} = - \int_\Omega u \varphi_{x_i}, \tag{1.2}$$

since φ is 0 on the boundary. Consequentially, for any function $u \in C^k(\Omega)$ and multi-index $\alpha \in \mathbb{N}^d$ with $|\alpha| \leq k$, we have

$$\int_\Omega D^\alpha u \varphi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \varphi, \tag{1.3}$$

using the fact that $D^\beta \varphi$ has compact support as well, for every $\beta \in \mathbb{N}^d$ with $|\beta| \leq k$. Note that for this last expression to make sense, the function $D^\alpha u$ does not necessarily have to be continuous. In fact, we can completely remove the hypothesis that u is k times continuously differentiable, and define the *weak derivative* of u as a function that satisfies the same integral Equation (1.3) for all test functions $\varphi \in C_c^\infty(\Omega)$.

Definition 1.10
Let $u, v \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}^d$. We say that v is the *α-th weak derivative* of u if

$$\int_\Omega v \varphi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \varphi \tag{1.4}$$

for every test function $\varphi \in C_c^\infty(\Omega)$.

We often use the same notation for weak derivatives as we do for regular derivatives. When $D^\alpha u$ is used, we should carefully think about which derivative is meant here. This notion extends to any differential operator that can classically be written in terms of partial derivatives, like the gradient ∇ or Laplacian $\Delta := \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$. To be able to talk about *the* α-th weak derivative, we show that weak derivatives are unique.

Proposition 1.11

If the α-th weak derivative of u exists, it is unique up to a set of measure zero.

The proof of this proposition is a good example of how the Fundamental Lemma of the Calculus of Variations can be used.

Proof. Suppose that v and w are two α-th weak derivatives of u. Then for every $\varphi \in C_c^\infty(\Omega)$,

$$\int_\Omega v\varphi = (-1)^{|\alpha|} \int_\Omega u D^\alpha \varphi = \int_\Omega w\varphi,$$

which gives

$$\int_\Omega (v - w)\varphi = 0$$

for every $\varphi \in C_c^\infty(\Omega)$. We can now use the Lemma 1.7 to see that $v - w = 0$ almost everywhere, which means that $v = w$ almost everywhere, completing the proof. □

With this new notion of weak differentiability, we can define new function spaces. These spaces are fundamental in the analysis of PDEs and will be used a lot in this thesis.

Definition 1.12

For $k \in \mathbb{N}$ and $p \in [1, \infty)$, the *Sobolev space of order* k, denoted $W^{k,p}(\Omega)$, is the space of functions in $L^p(\Omega)$ for which the α-th weak derivative exists and is in $L^p(\Omega)$, for all $|\alpha| \leq k$:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega),\ |\alpha| \leq k\}.$$

It is equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}.$$

The space $W^{k,p}(\Omega)$ complete, and thus forms a Banach space.

Recall that $L^p(\Omega)$ is a Banach space for every $p \in [1, \infty]$ and additionally forms a Hilbert space for $p = 2$. This holds for Sobolev spaces as well. We write $H^k(\Omega) := W^{k,2}(\Omega)$ and equip it with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)} ,$$

forming a Hilbert space. We write $H^k(\Omega)^*$ for its dual space and $\langle u, v \rangle$ for the pairing between $H^k(\Omega)^*$ and $H^k(\Omega)$, i.e.

$$\langle u, v \rangle := u(v)$$

for all $u \in H^k(\Omega)^*$ and $v \in H^k(\Omega)$.

In this thesis, we will be studying a bulk-surface reaction-diffusion model. This means that a function v defined on Ω will be reacting with some functions defined on a subset of the boundary $\partial\Omega$. These reactions will depend on the values of v on this boundary, but by definition, $H^1(\Omega)$ is a subset of $L^2(\Omega)$, so the functions that make up this Sobolev space are only defined up to a set of measure zero. The problem lies in the fact that the boundary *is* a set of measure zero, so we can generally not say anything about the values that are attained on it yet. We can, however, consider the case where we have a continuously differentiable function $u \in C^1(\overline{\Omega})$, for which the values on $\partial\Omega$ are certainly defined. We can also see u as a function in $H^1(\Omega) \cap C(\overline{\Omega})$, and define $v|_{\partial\Omega} := u|_{\partial\Omega}$ for every $v \in H^1(\Omega) \cap C(\overline{\Omega})$ that is equal to u almost everywhere in Ω. To further develop this idea, we first state the following property, proved in [Eva10].

Proposition 1.13 (Approximation by smooth functions)
If the boundary $\partial\Omega$ is C^1 and $1 \leq p < \infty$, the set of all smooth functions $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.

More precisely, for any function $u \in W^{k,p}(\Omega)$, there is a sequence of smooth functions defined on the closure of Ω, $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$, such that

$$\|u - u_n\|_{W^{k,p}(\Omega)} \to 0 \qquad \text{as } n \to \infty.$$

This allows us to extend our knowledge of functions defined on the boundary to functions in $W^{k,p}(\Omega)$ in general.

Theorem 1.14 (Trace Theorem)
If the boundary $\partial\Omega$ is C^1 and $1 \leq p < \infty$, there is a bounded linear operator $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$ such that

(i) $\gamma u = u|_{\partial\Omega}$ for any $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$ and

(ii) $\|\gamma u\|_{L^p(\partial\Omega)} \leq C_\gamma \|u\|_{W^{1,p}(\Omega)}$,

where the constant $C_\gamma > 0$ only depends on p and Ω.

A full proof of this theorem can be found in [Eva10], but we will describe the general outline here.

Proof. As we previously discussed, we can start by defining $\gamma u := u|_{\partial\Omega}$ for any $u \in C^1(\overline{\Omega})$. The first step in extending this definition is to use that the boundary is Lipschitz to locally flatten it and prove Item (ii) for $u \in C^1(\overline{\Omega})$ in a neighbourhood of some point $x \in \partial\Omega$. We can then use the fact that Ω is bounded to see that $\partial\Omega$ is compact, which allows us to get a finite number of points $x_1, \dots, x_n \in \partial\Omega$ with corresponding constants $C_1, \dots, C_n$, their neighbourhoods covering $\partial\Omega$. Combining the results for each x_i, we can conclude Item (ii) for the entirety of $\partial\Omega$.

For the general case of $u \in W^{1,p}(\Omega)$, we can use Proposition 1.13 to get an approximation for u of smooth functions $(u_n)_{n\in\mathbb{N}} \subseteq C^\infty(\overline{\Omega})$. The first part of the proof then implies that

$$\|\gamma u_n - \gamma u_m\|_{L^2(\partial\Omega)} = \|\gamma(u_n - u_m)\|_{L^2(\partial\Omega)} \leq C\|u_n - u_m\|_{W^{k,p}(\Omega)},$$

which means that $(\gamma u_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $L^2(\partial\Omega)$. We have used that γ is linear on $C^1(\overline{\Omega})$, since it simply acts as a restriction there. Using the fact that $L^2(\partial\Omega)$ is a Hilbert space, we can define

$$\gamma u := \lim_{n\to\infty} \gamma u_n = \lim_{n\to\infty} u_n|_{\partial\Omega},$$

which satisfies the same bound.

The proof is completed by noting that if $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$, then there are smooth functions $(u_n)_{n\in\mathbb{N}} \subseteq C^\infty(\overline{\Omega})$ converging to u uniformly on $\overline{\Omega}$, so $\gamma u = u|_{\partial\Omega}$. ☐

Definition 1.15
This operator $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$ is referred to as the *trace operator*. Moreover, γu is called the *trace* of u, and is sometimes denoted by $u|_{\partial\Omega}$ or just u.

1.4 Weak time derivatives

To incorporate time into these new notions of weak differentiability, we first develop the concept of L^p-spaces of functions that are not necessarily real- or complex-valued. Instead, let X be any Banach space and $T > 0$. A function $f : [0, T] \to X$ is then called *strongly measurable* if there is a sequence of simple functions $(f_n)_{n\in\mathbb{N}}$ that converges pointwise to f at almost every $t \in [0, T]$. Now for $p \in [1, \infty]$, the space $L^p(0, T; X)$ consists of all strongly measurable functions u such that $\|u\|_{L^p(0,T;X)} < \infty$, where

$$\|u\|_{L^p(0,T;X)} := \begin{cases} \left(\int_0^T \|u(t)\|_X^p \,\mathrm{d}t\right)^{1/p} & \text{if } p \in [1, \infty), \\ \operatorname{ess\,sup}_{t\,\in\,[0,T]} \|u(t)\|_X & \text{if } p = \infty. \end{cases}$$

These spaces are called *Bochner spaces*, and just as with traditional L^p-spaces, we can use them to define weak derivatives in time.

Definition 1.16
We say that $v \in L^1(0, T; X)$ is the *weak time derivative* of $u \in L^1(0, T; X)$, written $v = u'$ or

$v = u_t$, when

$$\int_0^T u(t)\varphi'(t)\,\mathrm{d}t = -\int_0^T v(t)\varphi(t)\,\mathrm{d}t$$

for all $\varphi \in C_c^\infty(0,T)$.

These weakly differentiable functions make up a different kind of Sobolev space, dependent on time.

Definition 1.17
The Sobolev space $W^{1,p}(0,T;X)$ consists of all functions $u \in L^p(0,T;X)$ such that u_t exists in the weak sense and belongs to $L^p(0,T;X)$. This space is equipped with the norm

$$\|u\|_{W^{1,p}(0,T;X)} := \begin{cases} \left(\int_0^T \|u(t)\|_X^p + \|u_t(t)\|_X^p \,\mathrm{d}t\right)^{1/p} & \text{if } p \in [1,\infty), \\ \operatorname{ess\,sup}_{t\,\in\,[0,T]} \left(\|u(t)\|_X + \|u_t(t)\|_X\right) & \text{if } p = \infty, \end{cases}$$

with which it forms a Banach space. We write $H^1(0,T;X) := W^{1,2}(0,T;X)$.

Lastly, $C([0,T];X)$ denotes the space of all continuous functions $u : [0,T] \to X$ with

$$\|u\|_{C([0,T];X)} := \max_{t\in[0,T]} \|u(t)\|_X < \infty.$$

We can take X to be, for instance, $L^2(\Omega)$. Then formally, $C([0,T];L^2(\Omega))$ consists of continuous functions u from the interval time [0, T] to $L^2(\Omega)$. For a more intuitive interpretation, you might feel inclined to interpret them as the corresponding functions $\widetilde{u}$ from $[0,T] \times \Omega$ to $\mathbb{R}$, given by $\widetilde{u}(t,x) = u(t)(x)$. We do need to be careful when we use this interpretation, though, as the notation might make it seem like these functions $\widetilde{u}$ are continuous in time, while this is generally not the case. Continuity in the L^2-norm gives that small changes in time produce small changes in overall spatial profile, but not necessarily at an individual point $x \in \Omega$.

There is an important connection between these two new spaces, which will be very useful in our later arguments for the existence and uniqueness of solutions to our model. It is a special case of a result known as the Lions-Magenes lemma, and was originally discussed in [LM72]. A more modern version is presented in [Eva10], though this is specified to a case assuming the function u vanishes on the boundary, something which is not needed for the proof.

Theorem 1.18 (Lions-Magenes)
Suppose that $u \in L^2(0,T;H^1(\Omega))$ with $u_t \in L^2(0,T;H^1(\Omega)^*)$. Then

(i) $u \in C([0,T];L^2(\Omega))$ after possibly being redefined on a set of measure zero,

(ii) the mapping $t \mapsto \|u(t)\|^2_{L^2(\Omega)}$ is absolutely continuous with derivative

$$\frac{\mathrm{d}}{\mathrm{d}t} \|u(t)\|^2_{L^2(\Omega)} = 2 \langle u_t(t), u(t) \rangle$$

for almost every $t \in [0, T]$, and

(iii) we have the estimate

$$\|u(t)\|_{C([0,T],L^2(\Omega))} \leq C \left(\|u\|_{L^2(0,T;H^1(\Omega))} + \|u_t\|_{L^2(0,T;H^1(\Omega)^*)} \right),$$

the constant C only depending on T.

We now prove a property of the trace operator that will be essential to our study of bulk-surface equations.

Proposition 1.19

If $v \in L^2(0, T; H^1(\Omega))$, then the mapping $t \mapsto \gamma(v(t))$ is in $L^2(0, T; L^2(\Gamma))$. Consequentially, for almost every $x \in \Gamma$, the function $t \mapsto \gamma(v(t))(x)$ is in $L^2(0, T)$.

Proof. We begin by showing that the function $t \mapsto \gamma(v(t))$ is strongly measurable. We know that $v : [0, T] \to H^1(\Omega)$ is strongly measurable, so there are simple functions $v_n : [0, T] \to H^1(\Omega)$, $n \in \mathbb{N}$ such that $v_n \to v$ pointwise for almost every $t \in [0, T]$. Since γ is a bounded linear operator due to Theorem 1.14, we can take the limit inside and see that

$$\lim_{n\to\infty} \gamma(v_n(t)) = \gamma \left(\lim_{n\to\infty} v_n(t) \right) = \gamma(v(t))$$

for almost every $t \in [0, T]$. Since $t \mapsto \gamma(v_n(t))$ is a simple function for every $n \in \mathbb{N}$, this shows that $t \mapsto \gamma(v(t))$ is strongly measurable.

We can bound its norm by

$$\|\gamma v\|^2_{L^2(0,T;L^2(\Gamma))} = \int_0^T \|\gamma(v(t))\|^2_{L^2(\Gamma)} \,\mathrm{d}t \leq C_\gamma^2 \int_0^T \|v(t)\|^2_{H^1(\Omega)} \,\mathrm{d}t = C_\gamma^2 \|v\|^2_{L^2(0,T;H^1(\Omega))} < \infty,$$

showing that indeed, $t \mapsto \gamma(v(t))$ is in $L^2(0, T; L^2(\Gamma))$.

Now by [HvNVW16], $L^2(0, T; L^2(\Gamma))$ is isometrically isomorphic to $L^2((0, T) \times \Gamma)$, so we have a representative of γv in $L^2((0, T) \times \Gamma)$. Then by Fubini's theorem we can say that for almost every x, $t \mapsto \gamma(v(t))(x)$ is in $L^2(0, T)$. ☐

1.5 Variational forms

Using the notion of weak derivatives, we can weaken the requirements for solutions to a PDE. Consider, for instance, the heat equation in $(0, T) \times \Omega$, where $\Omega \subseteq \mathbb{R}^m$ is open and bounded with a smooth boundary, with Neumann boundary conditions,

$$\begin{aligned} u_t &= \Delta u + f \quad &&\text{in } (0,T)\times\Omega,\\ \frac{\partial u}{\partial \nu} &= g &&\text{on } [0,T)\times\partial\Omega, \end{aligned} \tag{1.5}$$

where f and g are continuous functions on $(0,T)\times\Omega$ and $[0,T)\times\partial\Omega$ respectively. For a function u to be a classical solution, we require u to be of class $C^{1,2}((0,T)\times\Omega)\cap C^0([0,T]\times\overline{\Omega})$, i.e. continuously differentiable with respect to t, twice continuously differentiable with respect to x and continuously extendable to $[0,T]\times\overline{\Omega}$. We can multiply the heat equation by a smooth function $\varphi\in C^\infty(\Omega)$, integrate over Ω and integrate by parts to derive

$$\int_\Omega u_t\varphi = \int_\Omega \Delta u\varphi + \int_\Omega f\varphi = \int_{\partial\Omega}\frac{\partial u}{\partial\nu}\varphi \,\mathrm{d}S - \int_\Omega \nabla u\cdot\nabla\varphi + \int_\Omega f\varphi = \int_{\partial\Omega} g\varphi\,\mathrm{d}S - \int_\Omega \nabla u\cdot\nabla\varphi + \int_\Omega f\varphi.$$

We can also write this as

$$(u_t,\varphi)_{L^2(\Omega)} + (\nabla u,\nabla\varphi)_{L^2(\Omega;\mathbb{R}^d)} = (g,\varphi)_{L^2(\Gamma)} + (f,\varphi)_{L^2(\Omega)}\,, \tag{1.6}$$

and by Proposition 1.13, we can show that this should also hold for every $\varphi\in H^1(\Omega)$. Now note that for Equation (1.6) to be well-defined, we do not need u to be twice continuously differentiable anymore. In fact, we do not even need ∇u to be continuous. Instead, square-integrability of both ∇u and u_t is sufficient. This means we can substantially weaken the requirements for u, and instead ask if there exists a function $u\in H^1(0,T;H^1(\Omega))$ that satisfies Equation (1.6) for every $\varphi\in H^1(\Omega)$ and almost every $t\in[0,T]$.

Chapter 2

A bulk-surface model

In this chapter, we will discuss a bulk-surface model that describes the interactions of ligands and cell receptors. We will give a proof of the existence and uniqueness of weak solutions to this model, as well as discuss some of the qualitative properties of such solutions. The proof will rely heavily on the definitions and results introduced in Chapter 1, and will be based on an application of Banach's fixed-point theorem.

2.1 Derivation of the model

Consider an open and bounded set $U \subseteq \mathbb{R}^d$ with C^1 boundary and $T > 0$. This can be interpreted as the microscopic part of the body we are investigating. We identify the cells in this part of the body by an open set $C \subseteq U$, also with C^1 boundary, and the extracellular space by $\Omega := U \setminus \overline{C}$. We require that the closure $\overline{C}$ of C is also completely contained in U, i.e. no cells should touch the boundary of our domain. In particular, this implies that the boundary $\partial\Omega$ of the extracellular space can be written as the disjoint union of the boundary of our region of investigation ∂U and the cell membranes ∂C, which we denote by Γ_e and Γ, respectively. We use the function $u : [0, T] \times \Gamma \to \mathbb{R}$ to represent the concentration of free receptors at a given time and location on a cell membrane. Similarly, $u_b : [0, T] \times \Gamma \to \mathbb{R}$ represents the concentration of bounded receptors, and $v : [0, T] \times \overline{\Omega} \to \mathbb{R}$ represents the concentration of ligands in $\overline{\Omega}$, see Figure 2.1.

We derive a model that is similar to the ones used in [KMI14].The ligands can move freely through the extracellular space. This can be modelled by the standard diffusion (or heat) equation. To simplify the setting, we assume the individual receptors will be fixed in their position on the cell membrane.

When a ligand binds to a free receptor, the concentrations of free receptors and ligands decrease, and the concentration of bounded receptors at that position increases accordingly. This reaction is described by $k_{\text{on}}u^2v$ for some $k_{\text{on}} > 0$. After some time, the bounded receptor has done its task, passing the signal through to the interior of the cell. It releases the ligand and is now available to bind to a new ligand again. The concentration of bounded receptors decreases, and the concentration of free receptors and ligands increase accordingly. The rate of this dissociation is described by $k_{\text{off}}u_b$, where $k_{\text{off}} > 0$.

Finally, two reactions that happen constantly are production and decay. To keep intercellular communication possible, cells produce ligands and receptors that enter the domain Ω through the

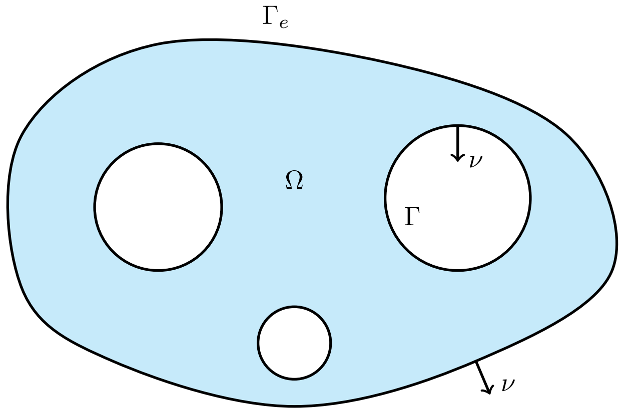

Figure 2.1: A sketch of a possible domain in the case that $d = 2$. The blue background represents Ω and the two kinds of boundaries are indicated, along with the outward unit normal vector ν.

cell membrane. We assume the production of ligands and free receptors mainly happens at constant rates $P_u > 0$ and $P_v > 0$, but is also upregulated by the concentration of bounded receptors, which is taken into account by αu_b for some upregulation coefficient $\alpha > 0$. The ligands, free receptors and receptor complexes decay at constant rates $\delta_v > 0$, $\delta_u > 0$ and $\delta_{u_b} > 0$, respectively.

We can describe the dynamics of ligands and receptors by a system of coupled ODEs given by

$$\begin{aligned} u_t &= P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 v + 2k_{\text{off}} u_b && \text{on } (0,T) \times \Gamma, \\ u_{b,t} &= -\delta_{u_b} u_b + k_{\text{on}} u^2 v - k_{\text{off}} u_b && \text{on } (0,T) \times \Gamma. \end{aligned} \tag{2.1}$$

Note that even though this is a system of two equations, these are actually uncountably many ODEs. For every boundary point $x \in \Gamma$, we have one ODE for the function $t \mapsto u(t,x)$ and one for the function $t \mapsto u_b(t,x)$.

Denote the diffusion coefficient by $D_v > 0$ and the outward unit normal vector on $\partial\Omega$ by ν. Since the dynamics of the ligands include diffusion in Ω and reaction on Γ, they can be described by the PDE and boundary conditions given by

$$\begin{aligned} v_t &= D_v \Delta v - \delta_v v && \text{in } (0,T) \times \Omega, \\ D_v \frac{\partial v}{\partial \nu} &= P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b && \text{on } [0,T) \times \Gamma, \\ D_v \frac{\partial v}{\partial \nu} &= 0 && \text{on } [0,T) \times \Gamma_e. \end{aligned} \tag{2.2}$$

Additionally, we need to specify initial conditions for the concentrations of ligands and receptors. That is, the solutions should satisfy

$$\begin{aligned} u(0,\cdot\,) &= u_0 && \text{on } \Gamma, \\ u_b(0,\cdot\,) &= u_{b,0} && \text{on } \Gamma, \\ v(0,\cdot\,) &= v_0 && \text{on } \Omega, \end{aligned} \tag{2.3}$$

for some given non-negative functions $u_0, u_{b,0} : \Gamma \to \mathbb{R}$ and $v_0 : \Omega \to \mathbb{R}$.

The remainder of this chapter will be dedicated to studying the system of Equations (2.1) to (2.3).

2.2 Weak formulation

Before we start proving the existence of solutions to this model, we must discuss what class of solutions we are considering. Specifically, we will derive the weak formulations of Equations (2.1) to (2.3) and discuss which function spaces are suitable.

We first consider the PDE, proceeding similarly to Section 1.5. To state the weak formulation, we assume u, u_b and v are smooth solutions to Equations (2.1) to (2.3). We multiply the PDE by a smooth function $\varphi \in C^\infty(\Omega)$ and integrate over Ω, giving

$$\begin{aligned}\int_\Omega v_t\varphi &= D_v\int_\Omega \Delta v\varphi - \delta_v\int_\Omega v\varphi \\ &= D_v\int_{\partial\Omega}\frac{\partial v}{\partial\nu}\varphi\,\mathrm{d}S - D_v\int_\Omega \nabla v\cdot\nabla\varphi - \delta_v\int_\Omega v\varphi \\ &= D_v\int_\Gamma\frac{\partial v}{\partial\nu}\varphi\,\mathrm{d}S + D_v\int_{\Gamma_e}\frac{\partial v}{\partial\nu}\varphi\,\mathrm{d}S - D_v\int_\Omega \nabla v\cdot\nabla\varphi - \delta_v\int_\Omega v\varphi \\ &= P_v\int_\Gamma\varphi\,\mathrm{d}S - k_{\text{on}}\int_\Gamma u^2v\varphi\,\mathrm{d}S + k_{\text{off}}\int_\Gamma u_b\varphi\,\mathrm{d}S - D_v\int_\Omega\nabla v\cdot\nabla\varphi - \delta_v\int_\Omega v\varphi,\end{aligned}$$

for all $t \in (0,T)$, which we can write using inner product notation as

$$(v_t,\varphi)_{L^2(\Omega)} = (P_v,\varphi)_{L^2(\Gamma)} - k_{\text{on}}\left(u^2v,\varphi\right)_{L^2(\Gamma)} + k_{\text{off}}\left(u_b,\varphi\right)_{L^2(\Gamma)} - D_v\left(\nabla v,\nabla\varphi\right)_{L^2(\Omega;\mathbb{R}^d)} - \delta_v\left(v,\varphi\right)_{L^2(\Omega)}.$$

We can substantially weaken the regularity conditions on v while keeping the terms in this equation well-defined. Both v and ∇v should be square integrable in space, i.e. we would need $v \in L^2(0,T;H^1(\Omega))$. Instead of assuming the same regularity for v_t, we can generalise even further and assume that $v_t \in L^2(0,T;H^1(\Omega)^*)$. This uses the fact that $L^2(\Omega)$ is identified with a subspace of $H^1(\Omega)^*$ by identifying a function $f \in L^2(\Omega)$ with a functional $f^* \in H^1(\Omega)^*$ defined by

$$\langle f^*,\varphi\rangle := (f,\varphi)_{L^2(\Omega)}, \qquad \varphi \in H^1(\Omega).$$

As a result, we can rewrite

$$\langle v_t,\varphi\rangle = (P_v,\varphi)_{L^2(\Gamma)} - k_{\text{on}}\left(u^2v,\varphi\right)_{L^2(\Gamma)} + k_{\text{off}}\left(u_b,\varphi\right)_{L^2(\Gamma)} - D_v\left(\nabla v,\nabla\varphi\right)_{L^2(\Omega;\mathbb{R}^d)} - \delta_v\left(v,\varphi\right)_{L^2(\Omega)}. \tag{2.4}$$

Note that this in fact requires higher integrability of the solution u. Otherwise, u^2v might not be square integrable, so the term $(u^2v,\varphi)_{L^2(\Gamma)}$ would not be well-defined. Therefore, we will consider solutions u that are essentially bounded. Using Proposition 1.13, we can show that Equation (2.4) also holds for more general functions $\varphi \in H^1(\Omega)$. Additionally, since we used integration with respect to space in this derivation, this expression only depends on time. For these reasons, a weak solution of Equation (2.2) is a function v in a suitable Sobolev space that satisfies Equation (2.4) for almost every $t \in (0,T)$ and for all $\varphi \in H^1(\Omega)$.

For the ODEs, we want to use Theorem 1.9, i.e. we will be looking for solutions of the system in the sense of Carathéodory on $[0,T]$ for fixed points on Γ. To this end, we assume a solution $v \in \mathcal{Y}$ of the PDE exists. We fix $x \in \Gamma$ and look for an absolutely continuous function $w := (u(\cdot, x), u_b(\cdot, x))$ such that

$$w'(t) = F(t, w(t)) \tag{2.5}$$

almost everywhere, where $F : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$ is given by

$$F(t, (u, u_b)) = \begin{pmatrix} P_u + \alpha u_b - \delta_u u - 2k_{on} u^2 \gamma v(t, x) + 2k_{\text{off}} u_b \\ -\delta_{u_b} u_b + k_{on} u^2 \gamma v(t, x) - k_{\text{off}} u_b \end{pmatrix}, \tag{2.6}$$

where γv denotes the trace of v as discussed in Section 1.3. Note that this definition of F takes into account that we assumed the existence of v. We will justify this in the next section, when we separate the problems for the ODEs and the PDE. Also note that this definition depends on the fact that x is assumed to be fixed. Thus for every $x \in \Gamma$, this defines a different ODE.

We can now combine these two formulations and start discussing the existence and uniqueness of their solutions.

Definition 2.1
The functions $u, u_b \in H^1(0, T; L^2(\Gamma))$ and $v \in L^2(0, T; H^1(\Omega))$ with $v_t \in L^2(0, T; H^1(\Omega)^*)$ are *weak solutions* to Equations (2.1) to (2.3) if v satisfies Equation (2.4) for almost every $t \in [0, T]$ and $\varphi \in H^1(\Omega)$ and

$$w(t) = (u(t)(x), u_b(t)(x))$$

satisfies Equation (2.5) for almost every $t \in [0, T]$ and $x \in \Gamma$.

2.3 Existence and uniqueness

Our goal in this section is to prove the following theorem.

Theorem 2.2 (Existence and uniqueness of weak solutions)
Let the initial conditions $u_0, u_{b,0} \in L^2(\partial\Omega)$ and $v_0 \in L^2(\Omega)$ be essentially bounded and non-negative. Then there exist unique weak solutions $u, u_b \in H^1(0, T; L^2(\Gamma))$ and $v \in L^2(0, T; H^1(\Omega))$ with $v_t \in L^2(0, T; H^1(\Omega)^*)$ to Equations (2.1) to (2.3). Moreover, these solutions are also non-negative and bounded.

As previously mentioned, the proof will be based on an application of Banach's fixed-point theorem. The proof for our model follows the same general idea as the proof of the Picard-Lindelöf theorem proving the existence and uniqueness for ODEs, but we need to adjust the arguments and function spaces to handle our situation of a coupled bulk-surface system. The idea is to consider the problem at a smaller time interval $[0, \widetilde{T}]$ for some $\widetilde{T} \leq T$ and "freeze" the function v in the ODEs so we can use the standard existence and uniqueness theory for ODEs to get solutions u and u_b to this problem. The PDE we treat similarly, freezing the functions u and u_b. Having a solution of this PDE, we can "update" the frozen terms in the ODE. This produces an operator $\mathcal{T}$ that maps any $v \in \mathcal{Y}$ to another $\widetilde{v} \in \mathcal{Y}$, where $\mathcal{Y}$ is the space defined by

$$\mathcal{Y} := \left\{ v \in L^2(0, \widetilde{T}; H^1(\Omega)) \;\middle|\; v_t \in L^2(0, \widetilde{T}; H^1(\Omega)^*) \text{ and } 0 \leq v \leq M_v(T) \text{ almost everywhere} \right\},$$

which we view as a closed subset of $L^2(0, \widetilde{T}; L^2(\Omega))$. We will derive the exact value of $M_v(T)$ later. Defining this operator, we will need another function space $\mathcal{X}$ given by

$$\mathcal{X} := \left\{ (u, u_b) \in \left(H^1(0, \widetilde{T}; L^2(\Gamma)) \right)^2 \;\middle|\; 0 \leq u \leq M_u(T) \text{ and } 0 \leq u_b \leq M_{u_b}(T) \text{ almost everywhere} \right\},$$

which we view as a closed subset of the space $(L^2(0, \widetilde{T}; L^2(\Gamma)))^2$. Again, the exact values of $M_u(T)$ and $M_{u_b}(T)$ will be derived later. If we can then show that this operator is a contraction, Banach's fixed-point theorem will guarantee the existence and uniqueness of a fixed point $v^* \in \mathcal{Y}$, which will be a local solution to the PDE. We will then show how we can stitch together local solutions to get a unique solution for every final time T.

Let us first make precise what we mean by "freezing" the function v in the ODE system. Fix $\widetilde{T} \leq T$ and $v^* \in \mathcal{Y}$ and consider the modified system of ODEs given by

$$\begin{aligned} u_t &= P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 \gamma v^* + 2k_{\text{off}} u_b && \text{on } (0, \widetilde{T}) \times \Gamma, \\ u_{b,t} &= -\delta_{u_b} u_b + k_{\text{on}} u^2 \gamma v^* - k_{\text{off}} u_b && \text{on } (0, \widetilde{T}) \times \Gamma, \\ u(0) &= u_0, \qquad u_b(0) = u_{b,0} && \text{on } \Gamma. \end{aligned} \tag{2.7}$$

These ODEs are now *decoupled* from the PDE, so we can investigate the existence and uniqueness of solutions to this system without simultaneously having to prove that v is a solution of the PDE.

Lemma 2.3 (Existence and uniqueness for the decoupled ODEs)
Let the initial conditions $u_0, u_{b,0} \in L^2(\Gamma)$ be essentially bounded and non-negative and fix $v^* \in \mathcal{Y}$. Then there exist unique weak solutions $(u, u_b) \in \mathcal{X}$ to Equation (2.7) satisfying these initial conditions.

Proof. Fix x and define w and $F : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$ as in Equations (2.5) and (2.6), where v is replaced by v^*. Since the components of F are polynomial in w, the function F is certainly continuous in w. Additionally, by the regularity of γv^* which we showed in Proposition 1.19, we know that F is measurable in t. This also implies that for any compact subset of $\mathbb{R} \times \mathbb{R}^2$, we can bound F by some time-dependent integrable function m. In particular, this subset can be taken to be a rectangle[1] as in the hypothesis of Theorem 1.9, so let R be such a rectangle around the initial time $t = 0$ and initial data $(u_0, u_{b,0})$. The hypotheses of Theorem 1.9 are then fully satisfied, giving us a local solution $w : [0, \beta] \to \mathbb{R}^2$ for some $\beta > 0$, with $w(0) = (u_0, u_{b,0})$.

We show that this local solution is unique. Suppose we have two such local solutions, w_1 and w_2. Since these solutions are absolutely continuous, they are bounded on $[0, \beta]$, so we can take a compact set $K \subseteq \mathbb{R}^2$ that contains both their trajectories. For any $t \in [0, \beta]$ and $\overline{w} = (\overline{u}, \overline{u}_b), \widetilde{w} =$

[1] On first sight, it can seem strange to call this type of set a rectangle instead of a cylinder. This name comes from the fact that in our main source [CL55], the theorem is proved for the case of $n = 1$.

$(\widetilde{u}, \widetilde{u}_b) \in K$, F satisfies

$$\begin{aligned}
|F(t,\overline{w}) - F(t,\widetilde{w})|^2 &= \left| \begin{pmatrix} -2k_{\text{on}}\left(\overline{u}^2 - \widetilde{u}^2\right)\gamma v^*(t,x) + 2k_{\text{off}}\left(\overline{u}_b - \widetilde{u}_b\right) + \alpha\left(\overline{u}_b - \widetilde{u}_b\right) - \delta_u\left(\overline{u} - \widetilde{u}\right) \\ k_{\text{on}}\left(\overline{u}^2 - \widetilde{u}^2\right)\gamma v^*(t,x) - k_{\text{off}}\left(\overline{u}_b - \widetilde{u}_b\right) - \delta_{u_b}\left(\overline{u}_b - \widetilde{u}_b\right) \end{pmatrix} \right|^2 \\
&= \left| \begin{pmatrix} -\left(2k_{\text{on}}\left(\overline{u} + \widetilde{u}\right)\gamma v^*(t,x) + \delta_u\right)\left(\overline{u} - \widetilde{u}\right) & +\left(2k_{\text{off}} + \alpha\right)\left(\overline{u}_b - \widetilde{u}_b\right) \\ k_{\text{on}}\left(\overline{u} + \widetilde{u}\right)\gamma v^*(t,x)\left(\overline{u} - \widetilde{u}\right) & -\left(k_{\text{off}} + \delta_{u_b}\right)\left(\overline{u}_b - \widetilde{u}_b\right) \end{pmatrix} \right|^2 \\
&\leq 2\left| \begin{pmatrix} -2k_{\text{on}}\left(\overline{u} + \widetilde{u}\right)\gamma v^*(t,x) - \delta_u \\ k_{\text{on}}\left(\overline{u} + \widetilde{u}\right)\gamma v^*(t,x) \end{pmatrix} \left(\overline{u} - \widetilde{u}\right) \right|^2 + 2\left| \begin{pmatrix} 2k_{\text{off}} + \alpha \\ k_{\text{off}} + \delta_{u_b} \end{pmatrix} \left(\overline{u}_b - \widetilde{u}_b\right) \right|^2 \\
&\leq L_K(t)^2 \left(\left(\overline{u} - \widetilde{u}\right)^2 + \left(\overline{u}_b - \widetilde{u}_b\right)^2 \right) \leq L_K(t)^2 \left|\overline{w} - \widetilde{w}\right|^2,
\end{aligned}$$

where $L_K(t)$ is defined as

$$L_K(t) := \sqrt{2}\max\left\{ \left| \begin{pmatrix} 4k_{\text{on}}\operatorname{diam}(K)v^*(t,x) + \delta_u \\ 2k_{\text{on}}\operatorname{diam}(K)v^*(t,x) \end{pmatrix} \right|, \quad \left| \begin{pmatrix} 2k_{\text{off}} + \alpha \\ k_{\text{off}} + \delta_{u_b} \end{pmatrix} \right| \right\},$$

which is of L^2-regularity in t due to the essential boundedness of v^*. Now for the difference of the two solutions we obtain

$$\begin{aligned}
|w_1(t) - w_2(t)| &= \left| (u_0, u_{b,0}) + \int_0^t F(t, w_1(t))\,\mathrm{d}t - (u_0, u_{b,0}) - \int_0^t F(t, w_2(t))\,\mathrm{d}t \right| \\
&\leq \int_0^t |F(t, w_1(t)) - F(t, w_2(t))|\,\mathrm{d}t \leq \int_0^t L_K(t)\,|w_1(t) - w_2(t)|\,\mathrm{d}t
\end{aligned}$$

for every $t \in [0, \beta]$. Using Gronwall's lemma and the fact that $w_1(0) = w_2(0)$, we see that this implies that $|w_1(t) - w_2(t)| = 0$ for all $t \in [0, \beta]$, so $w_1 = w_2$.

To show that the components u and u_b of this solution w are non-negative, we multiply the equations of Equation (2.7) by $u_- := -\min(0, u)$ and $u_{b,-} := -\min(0, u_b)$, respectively, and obtain

$$\begin{aligned}
(u_{b,-}(t))^2 &= 2\int_0^t u_{b,-}(r)\frac{\mathrm{d}}{\mathrm{d}t}u_{b,-}(r)\,\mathrm{d}r = -2\int_0^t u_{b,-}(r)u_{b,t}(r)\,\mathrm{d}r \\
&= -2\int_0^t \left[-\delta_{u_b}u_b(r) + k_{\text{on}}u(r)^2\gamma v^*(r,x) - k_{\text{off}}u_b(r)\right]u_{b,-}(r)\,\mathrm{d}r \\
&= 2\int_0^t \left[-\delta_{u_b}\left(u_{b,-}(r)\right)^2 - k_{\text{on}}u^2(r)\gamma v^*(r,x)u_{b,-}(r) - k_{\text{off}}\left(u_{b,-}(r)\right)^2\right]\mathrm{d}r \leq 0,
\end{aligned}$$

so $u_{b,-}(t) = 0$ for every $t \in [0, \beta]$, showing that u_b must be non-negative for almost every $t \in [0, \beta]$.

Using this in the estimate for u, we get

$$\begin{aligned}(u_-(t))^2 &= 2\int_0^t u_-(r)\frac{\mathrm{d}}{\mathrm{d}t}u_-(r)\,\mathrm{d}r = -2\int_0^t u_-(r)u_t(r)\,\mathrm{d}r\\&= -2\int_0^t \left[P_u + \alpha u_b(r) - \delta_u u(r) - 2k_{\text{on}}u(r)^2\gamma v^*(r,x) + 2k_{\text{off}}u_b(r)\right]u_-(r)\,\mathrm{d}r\\&= 4k_{\text{on}}\int_0^t u(r)^2\gamma v^*(r,x)u_-(r)\,\mathrm{d}r\\&\quad - 2\int_0^t \left[P_u u_-(r) + \alpha u_b(r)u_-(r) + \delta_u u(r)^2 + 2k_{\text{off}}u_b(r)u_-(r)\right]\mathrm{d}r\\&\le 4k_{\text{on}}\int_0^t u(r)^2\gamma v^*(r,x)u_-(r)\,\mathrm{d}r.\end{aligned}$$

Applying Gronwall's lemma, this shows that $u_-(t) = 0$ for almost all $t \in [0, \beta]$, so $u \ge 0$.

We can use the non-negativity of u and u_b to show that these functions are bounded. Define $r := u + 2u_b$ and note that

$$r_t = u_t + 2u_{b,t} = P_u - \delta_u u + (\alpha - 2\delta_{u_b})u_b \le P_u + mr,$$

where $m := \max\left(\delta_u, \frac{1}{2}(\alpha - 2\delta_{u_b})\right)$. Then Gronwall's lemma gives

$$r(t) \le e^{mt}\left(r(0) + tP_u\right).$$

Now the non-negativity of both u and u_b and the essential boundedness of u_0 and $u_{b,0}$ allows us to bound the solutions independent of $\widetilde{T}$ by

$$\begin{aligned}u &\le e^{mT}\left(M + P_u\right) =: M_u(T) \qquad \text{and}\\ u_b &\le \frac{1}{2}M_u(T) \qquad\quad\;\; =: M_{u_b}(T),\end{aligned} \tag{2.8}$$

where $M = \operatorname{ess\,sup}(u_0 + 2u_{b,0})$ and $T \ge \widetilde{T}$ is the original final time, for almost every $x \in \Gamma$ and $t \in [0, \beta]$.

Now suppose we have solutions u and u_b on a maximum time interval $[0, \widetilde{\beta})$. Since we have a bound for u and u_b that is independent of $\widetilde{\beta}$, we can use the continuation argument from [CL55] and see that we have a solution for u and u_b on $[0, \widetilde{T}]$.

Now if we let x vary, we can see that u and u_b are both of L^2 regularity in space, because the bounds from Equation (2.8) are independent of space. Lastly, the fact that u and u_b are essentially bounded shows that u_t and $u_{b,t}$ are bounded, so $u, u_b \in H^1(0, \widetilde{T}; L^2(\Gamma))$. □

We can use a similar approach for the PDE in Equation (2.2). Fix $(u^*, u_b^*) \in \mathcal{X}$ and consider the modified weak formulation given by

$$\begin{aligned}\langle v_t, \varphi\rangle = (P_v, \varphi)_{L^2(\Gamma)} - k_{\text{on}}\left((u^*)^2\, v, \varphi\right)_{L^2(\Gamma)} + k_{\text{off}}\,(u_b^*, \varphi)_{L^2(\Gamma)}\\ - D_v\,(\nabla v, \nabla\varphi)_{L^2(\Omega;\mathbb{R}^d)} - \delta_v\,(v, \varphi)_{L^2(\Omega)}\,.\end{aligned} \tag{2.9}$$

for almost every $t \in [0, \widetilde{T}]$. This PDE is now decoupled from the ODEs, and we can prove the existence and uniqueness independently from the ODE system.

Lemma 2.4 (Existence and uniqueness for the decoupled PDE)
Let the initial condition $v_0 \in L^2(\Omega)$ be essentially bounded and non-negative and fix $(u^*, u_b^*) \in \mathcal{X}$. Then there exists a unique weak solution $v \in \mathcal{Y}$ to Equation (2.9) satisfying these initial conditions.

Proof. To prove the existence and uniqueness goes beyond the scope of this thesis. Hence, we assume that a unique weak solution v of the PDE exists and v is essentially bounded by a constant depending on T. This can be proven using Galerkin methods, as demonstrated in [Eva10]. Let v now be such a solution. To show that v is non-negative, we estimate the negative part of v. Let v_+ and v_- be defined as $v_+ := \max(0, v)$ and $v_- := -\min(0, v)$ respectively, such that $v = v_+ - v_-$. Then as discussed in [Eva10], we have $v_+, v_- \in L^2(0, \widetilde{T}; H^1(\Omega))$ and $(v_+)_t, (v_-)_t \in L^2(0, \widetilde{T}; H^1(\Omega)^*)$. Furthermore, $v_t v_- = -(v_-)_t v_-$, as well as $\nabla v \cdot \nabla v_- = -|\nabla v_-|^2$, so for almost every t we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|v_-\|^2_{L^2(\Omega)} = \langle (v_-)_t, v_-\rangle = -\langle v_t, v_-\rangle$$

due to Theorem 1.18. We use v_- as a test function and employ the fact that v is a solution to the PDE, which gives

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|v_-\|^2_{L^2(\Omega)} &= -\langle v_t, v_-\rangle = D_v \int_\Omega \nabla v \cdot \nabla v_- + \delta_v \int_\Omega v v_- - \int_\Gamma \left(P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b\right) v_- \,\mathrm{d}S \\
&= -D_v \|\nabla v_-\|^2_{L^2(\Omega;\mathbb{R}^d)} - \delta_v \|v_-\|^2_{L^2(\Omega)} - P_v \int_\Gamma v_- \,\mathrm{d}S + k_{\text{on}} \int_\Gamma u^2 v v_- \,\mathrm{d}S - k_{\text{off}} \int_\Gamma u_b v_- \,\mathrm{d}S \\
&= -D_v \|\nabla v_-\|^2_{L^2(\Omega;\mathbb{R}^d)} - \delta_v \|v_-\|^2_{L^2(\Omega)} - P_v \int_\Gamma v_- \,\mathrm{d}S - k_{\text{on}} \int_\Gamma u^2 (v_-)^2 \,\mathrm{d}S - k_{\text{off}} \int_\Gamma u_b v_- \,\mathrm{d}S \\
&\leq 0,
\end{aligned}$$

where we used the non-negativity of v_- and u_b in the last step. By Theorem 1.18, we know that v is continuous in time with respect to the L^2-norm. The only way for this to be achieved while also guaranteeing that the time derivative is non-positive, is by having

$$\|v_-\|_{L^2(\Omega)} = 0$$

for almost every t, which implies that v is non-negative. □

Using Lemmas 2.3 and 2.4, we can define two operators between $\mathcal{Y}$ and $\mathcal{X}$. One takes a function $v^* \in \mathcal{Y}$ and maps it to the unique solution $(u, u_b) \in \mathcal{X}$ of Equation (2.7). Let us refer to this operator as $\mathcal{T}_1 : \mathcal{Y} \to \mathcal{X}$. The other operator, $\mathcal{T}_2 : \mathcal{X} \to \mathcal{Y}$, takes two arbitrary functions $(u^*, u_b^*) \in \mathcal{X}$ and defines $\mathcal{T}_2(u^*, u_b^*)$ as the unique solution $v \in \mathcal{Y}$ to Equation (2.9). Combining these two operators, we define

$$\mathcal{T} := \mathcal{T}_2 \circ \mathcal{T}_1 : \mathcal{Y} \to \mathcal{Y}.$$

We want to show that this operator has a unique fixed point. If this is the case, we have a function $v \in \mathcal{Y}$ such that $\mathcal{T}(v) = v$, and let $(u, u_b) \in \mathcal{X}$ denote $(u, u_b) = \mathcal{T}_1(v)$. Given that then $\mathcal{T}_2(u, u_b) = v$, by definition this means that v solves Equation (2.4) for almost every $t \in (0, \widetilde{T})$, and $w = (u, u_b)$

solves Equation (2.5) for almost every $t \in (0, \widetilde{T})$ and $x \in \Gamma$, meaning that v, u and u_b are weak solutions of our model on the local time scale $(0, \widetilde{T})$. Conversely, if v, u and u_b are weak solutions to the model, then by the uniqueness of solutions in Lemmas 2.3 and 2.4, we must have that $\mathcal{T}_2(u, u_b) = v$ and $\mathcal{T}_1(v) = (u, u_b)$, and thus $\mathcal{T}(v) = v$.

It should be clear now how we will apply Banach's fixed-point theorem. If we prove that $\mathcal{T}$ is a contraction, we can guarantee the existence of a fixed point and therefore a weak solution. Since every weak solution of the model is also a fixed point, the theorem also grants us uniqueness of such solutions. To prove that $\mathcal{T}$ is a contraction, we first show this for $\mathcal{T}_2$ and $\mathcal{T}_1$ separately, after which we can easily combine the results.

Lemma 2.5

For $\widetilde{T} > 0$ small enough, $\mathcal{T}_2 : \mathcal{X} \to \mathcal{Y}$ is a contraction. That is, there is a $q \in [0, 1)$ such that

$$\|\mathcal{T}_2(\overline{u}, \overline{u}_b) - \mathcal{T}_2(\widetilde{u}, \widetilde{u}_b)\|^2_{L^2(0,\widetilde{T};L^2(\Omega))} \leq q \,\|(\overline{u}, \overline{u}_b) - (\widetilde{u}, \widetilde{u}_b)\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2}$$

for every $(\overline{u}, \overline{u}_b), (\widetilde{u}, \widetilde{u}_b) \in \mathcal{X}$.

Proof. Let $(\overline{u}, \overline{u}_b), (\widetilde{u}, \widetilde{u}_b) \in \mathcal{X}$ and define $\overline{v}, \widetilde{v} \in \mathcal{Y}$ by $\overline{v} := \mathcal{T}_2(\overline{u}, \overline{u}_b)$ and $\widetilde{v} := \mathcal{T}_2(\widetilde{u}, \widetilde{u}_b)$. We now denote their differences by

$$v := \overline{v} - \widetilde{v}, \qquad u := \overline{u} - \widetilde{u} \qquad \text{and} \qquad u_b := \overline{u}_b - \widetilde{u}_b,$$

such that

$$\|\mathcal{T}_2(\overline{u}, \overline{u}_b) - \mathcal{T}_2(\widetilde{u}, \widetilde{u}_b)\|^2_{L^2(0,\widetilde{T};L^2(\Omega))} = \|\overline{v} - \widetilde{v}\|^2_{L^2(0,\widetilde{T};L^2(\Omega))} = \|v\|^2_{L^2(0,\widetilde{T};L^2(\Omega))}.$$

We can estimate this norm using the weak form of the PDE. We use Theorem 1.18 to see that

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|v\|^2_{L^2(\Omega)} = \langle v_t, v\rangle = -D_v\int_\Omega |\nabla v|^2 - \delta_v\int_\Omega v^2 + \int_\Gamma \left(-k_{\text{on}}\overline{u}^2\overline{v} + k_{\text{off}}\overline{u}_b + k_{\text{on}}\widetilde{u}^2\widetilde{v} - k_{\text{off}}\widetilde{u}_b\right) v \,\mathrm{d}S,$$

which after rearranging becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|v\|^2_{L^2(\Omega)} + D_v\|\nabla v\|^2_{L^2(\Omega;\mathbb{R}^d)} + \delta_v\|v\|^2_{L^2(\Omega)} = -k_{\text{on}}\int_\Gamma \left(\overline{u}^2\overline{v} - \widetilde{u}^2\widetilde{v}\right) v \,\mathrm{d}S + k_{\text{off}}\int_\Gamma u_b v \,\mathrm{d}S$$

for almost all t. We then use the identity

$$\overline{u}^2\overline{v} - \widetilde{u}^2\widetilde{v} = \overline{u}^2\overline{v} - \widetilde{u}^2\overline{v} + \widetilde{u}^2\overline{v} - \widetilde{u}^2\widetilde{v} = (\overline{u}^2 - \widetilde{u}^2)\overline{v} + \widetilde{u}^2 v = (\overline{u} + \widetilde{u})\overline{v}u + \widetilde{u}^2 v \tag{2.10}$$

to estimate

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|v\|^2_{L^2(\Omega)} &+ D_v\|\nabla v\|^2_{L^2(\Omega;\mathbb{R}^d)} + \delta_v\|v\|^2_{L^2(\Omega)} \\
&= -k_{\text{on}}\int_\Gamma (\overline{u} + \widetilde{u})\,\overline{v}uv \,\mathrm{d}S - k_{\text{on}}\int_\Gamma \widetilde{u}^2 v^2 \,\mathrm{d}S + k_{\text{off}}\int_\Gamma u_b v \,\mathrm{d}S \\
&\leq k_{\text{on}}\int_\Gamma |(\overline{u} + \widetilde{u})\overline{v}uv| \,\mathrm{d}S + k_{\text{off}}\int_\Gamma u_b v \,\mathrm{d}S \\
&\leq 2k_{\text{on}}M_u(T)M_v(T)\int_\Gamma |uv| \,\mathrm{d}S + k_{\text{off}}\int_\Gamma u_b v \,\mathrm{d}S.
\end{aligned}$$

The mixed terms can be separated using Young's inequality, giving

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\left\|v\right\|^2_{L^2(\Omega)} + D_v\left\|\nabla v\right\|^2_{L^2(\Omega;\mathbb{R}^d)} + \delta_v\left\|v\right\|^2_{L^2(\Omega)} \\
&\qquad\le 2k_{\text{on}}M_u(T)M_v(T)\left(A_\varepsilon\left\|u\right\|^2_{L^2(\Gamma)} + \varepsilon\left\|v\right\|^2_{L^2(\Gamma)}\right) + k_{\text{off}}\left(A_\varepsilon\left\|u_b\right\|^2_{L^2(\Gamma)} + \varepsilon\left\|v\right\|^2_{L^2(\Gamma)}\right) \\
&\qquad\le B_\varepsilon(T)\left\|u\right\|^2_{L^2(\Gamma)} + k_{\text{off}}A_\varepsilon\left\|u_b\right\|^2_{L^2(\Gamma)} + C(T)\varepsilon\left\|v\right\|^2_{L^2(\Gamma)} \\
&\qquad\le B_\varepsilon(T)\left\|u\right\|^2_{L^2(\Gamma)} + k_{\text{off}}A_\varepsilon\left\|u_b\right\|^2_{L^2(\Gamma)} + C_\gamma^2 C(T)\varepsilon\left\|v\right\|^2_{H^1(\Omega)}
\end{aligned}$$

for almost all t. Now let $\delta := \min(\delta_v, D_v)$ and integrate from 0 to $\widetilde{T}$ to get

$$\frac{1}{2}\left\|v(t)\right\|^2_{L^2(\Omega)} + (\delta - C_\gamma^2 C(T)\varepsilon)\left\|v\right\|^2_{L^2(0,\widetilde{T};H^1(\Omega))} \le B_\varepsilon(T)\left\|u\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + k_{\text{off}}A_\varepsilon\left\|u_b\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))}$$

for almost all t. If we now choose $\varepsilon < \frac{\delta}{C_\gamma^2 C(T)}$, we can estimate both the $L^2(0,\widetilde{T};H^1(\Omega))$-norm and the $L^2(0,\widetilde{T};L^2(\Omega))$-norm of v. The first will only be helpful later, but the latter concludes this proof. First, since $\delta - C_\gamma^2 C(T)\varepsilon > 0$, we have

$$\left\|v\right\|^2_{L^2(0,\widetilde{T};H^1(\Omega))} \le D_\varepsilon(T)\left(\left\|u\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \left\|u_b\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))}\right). \tag{2.11}$$

Second, we can integrate from 0 to $\widetilde{T}$ once more to get

$$\left\|v\right\|^2_{L^2(0,\widetilde{T};L^2(\Omega))} \le \widetilde{T}M_\varepsilon(T)\left(\left\|u\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \left\|u_b\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))}\right).$$

Now for $\widetilde{T} < \frac{1}{M_\varepsilon(T)}$, this coefficient is less than 1, so $\mathcal{T}_2$ will be a contraction. □

Lemma 2.6
For $\widetilde{T} > 0$ small enough, $\mathcal{T}_1 : \mathcal{Y} \to \mathcal{X}$ is a contraction. That is, there is a $q \in [0,1)$ such that

$$\left\|\mathcal{T}_1(\overline{v}) - \mathcal{T}_1(\widetilde{v})\right\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2} \le q\left\|\overline{v} - \widetilde{v}\right\|^2_{L^2(0,\widetilde{T};L^2(\Omega))}$$

for every $\overline{v}, \widetilde{v} \in \mathcal{Y}$.

Proof. Now let $\overline{v}, \widetilde{v} \in \mathcal{Y}$ and define $\overline{u}, \overline{u}_b, \widetilde{u}$ and $\widetilde{u}_b$ by $(\overline{u}, \overline{u}_b) := \mathcal{T}_1(\overline{v})$ and $(\widetilde{u}, \widetilde{u}_b) := \mathcal{T}_1(\widetilde{v})$. Once again, we denote their differences by

$$v := \overline{v} - \widetilde{v}, \qquad u := \overline{u} - \widetilde{u} \qquad \text{and} \qquad u_b := \overline{u}_b - \widetilde{u}_b,$$

such that

$$\begin{aligned}
\left\|\mathcal{T}_1(\overline{v}) - \mathcal{T}_1(\widetilde{v})\right\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2} &= \left\|(\overline{u}, \overline{u}_b) - (\widetilde{u}, \widetilde{u}_b)\right\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2} = \left\|(u, u_b)\right\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2} \\
&= \left\|u\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \left\|u_b\right\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))}.
\end{aligned}$$

We therefore seek to estimate these separate differences u and u_b in terms of v. Starting off with u, we see that it satisfies

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\|u\|^2_{L^2(\Gamma)} &= 2\int_\Gamma u_t u \,\mathrm{d}S = 2\int_\Gamma (\overline{u}_t - \widetilde{u}_t) u \,\mathrm{d}S \\
&= 2\int_\Gamma \big(\alpha(\overline{u}_b - \widetilde{u}_b) - \delta_u(\overline{u} - \widetilde{u}) - 2k_{\text{on}}(\overline{u}^2\overline{v} - \widetilde{u}^2\widetilde{v}) + 2k_{\text{off}}(\overline{u}_b - \widetilde{u}_b)\big)\, u \,\mathrm{d}S \\
&= 2(\alpha + 2k_{\text{off}})\int_\Gamma u u_b \,\mathrm{d}S - 2\delta_u \int_\Gamma u^2 \,\mathrm{d}S - 4k_{\text{on}}\int_\Gamma (\overline{u}^2\overline{v} - \widetilde{u}^2\widetilde{v}) u \,\mathrm{d}S
\end{aligned}$$

for almost every t. Here, the first step is justified because we know that u belongs to $H^1(0, \widetilde{T}; L^2(\Gamma))$. As we did in the proof of Lemma 2.5, we now use the identity in Equation (2.10) to conclude that

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\|u\|^2_{L^2(\Gamma)} &= 2(\alpha + 2k_{\text{off}})\int_\Gamma u u_b \,\mathrm{d}S - 2\delta_u \int_\Gamma u^2 \,\mathrm{d}S - \overbrace{4k_{\text{on}}\int_\Gamma (\overline{u} + \widetilde{u})\overline{v} u^2 \,\mathrm{d}S}^{\geq 0} - 4k_{\text{on}}\int_\Gamma \widetilde{u}^2 v u \,\mathrm{d}S \\
&\leq 2(\alpha + 2k_{\text{off}})\int_\Gamma u u_b \,\mathrm{d}S - 2\delta_u \int_\Gamma u^2 \,\mathrm{d}S + 4k_{\text{on}}\int_\Gamma \widetilde{u}^2\,|vu| \,\mathrm{d}S \\
&\leq 2(\alpha + 2k_{\text{off}})\int_\Gamma u u_b \,\mathrm{d}S - 2\delta_u \|u\|^2_{L^2(\Gamma)} + 4k_{\text{on}} M_u(T)^2 \int_\Gamma |vu| \,\mathrm{d}S.
\end{aligned}$$

To further separate the mixed terms, we use Young's inequality twice and obtain the estimate

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\|u\|^2_{L^2(\Gamma)} &\leq A_1(T)\|u\|^2_{L^2(\Gamma)} + A_2\|u_b\|^2_{L^2(\Gamma)} + \widetilde{B_1}(T)\|v\|^2_{L^2(\Gamma)}, \\
&\leq A_1(T)\|u\|^2_{L^2(\Gamma)} + A_2\|u_b\|^2_{L^2(\Gamma)} + B_1(T)\|v\|^2_{H^1(\Omega)},
\end{aligned}$$

where $A_1(T)$, A_2, $\widetilde{B_1}(T)$ and $B_1(T)$ are defined as

$$\begin{aligned}
&A_1(T) := -2\delta_u + \alpha + 2k_{\text{off}} + 2k_{\text{on}} M_u(T)^2, \qquad && A_2 := \alpha + 2k_{\text{off}} \\
&\widetilde{B_1}(T) := 2k_{\text{on}} M_u(T)^2 \qquad \text{and} && B_1(T) := C_\gamma^2 \widetilde{B_1}(T).
\end{aligned}$$

Using the same arguments, we get a similar estimate for $\frac{\mathrm{d}}{\mathrm{d}t}\|u_b\|^2_{L^2(\Gamma)}$, namely

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\|u_b\|^2_{L^2(\Gamma)} &= 2\int_\Gamma \left((\overline{u}_b)_t - (\widetilde{u}_b)_t\right) u_b \,\mathrm{d}S \\
&= -2\left(\delta_{u_b} + k_{\text{off}}\right)\int_\Gamma u_b^2 \,\mathrm{d}S + 2k_{\text{on}}\int_\Gamma \left(\overline{u}^2\overline{v} - \widetilde{u}^2\widetilde{v}\right) u_b \,\mathrm{d}S \\
&= -2\left(\delta_{u_b} + k_{\text{off}}\right)\int_\Gamma u_b^2 \,\mathrm{d}S + 2k_{\text{on}}\int_\Gamma (\overline{u} + \widetilde{u})\,\widetilde{v} u u_b \,\mathrm{d}S + 2k_{\text{on}}\int_\Gamma \widetilde{u}^2 v u_b \,\mathrm{d}S \\
&\leq -2\left(\delta_{u_b} + k_{\text{off}}\right)\int_\Gamma u_b^2 \,\mathrm{d}S + 2k_{\text{on}}\int_\Gamma (\overline{u} + \widetilde{u})\,\widetilde{v}\,|u u_b| \,\mathrm{d}S + 2k_{\text{on}}\int_\Gamma \widetilde{u}^2\,|v u_b| \,\mathrm{d}S \\
&\leq -2\left(\delta_{u_b} + k_{\text{off}}\right)\int_\Gamma u_b^2 \,\mathrm{d}S + 4k_{\text{on}} M_u(T) M_v(T)\int_\Gamma |u u_b| \,\mathrm{d}S + 2k_{\text{on}} M_u(T)^2 \int_\Gamma |v u_b| \,\mathrm{d}S
\end{aligned}$$

$$
\begin{aligned}
&\leq A_3(T)\,\|u_b\|^2_{L^2(\Gamma)} + A_4(T)\,\|u\|^2_{L^2(\Gamma)} + \widetilde{B_2}(T)\,\|v\|^2_{L^2(\Gamma)}\,, \\
&\leq A_3(T)\,\|u_b\|^2_{L^2(\Gamma)} + A_4(T)\,\|u\|^2_{L^2(\Gamma)} + B_2(T)\,\|v\|^2_{H^1(\Omega)}\,,
\end{aligned}
$$

where $A_3(T)$, $A_4(T)$, $\widetilde{B_2}(T)$ and $B_2(T)$ are given by

$$A_3(T) := 2k_{\text{on}}M_u(T)M_v(T) + k_{\text{on}}M_u(T)^2 - 2\delta_{u_b} - 2k_{\text{off}},$$

$$A_4(T) := 2k_{\text{on}}M_u(T)M_v(T), \qquad \widetilde{B_2}(T) := k_{\text{on}}M_u(T)^2 \qquad \text{and} \qquad B_2(T) := C_\gamma^2\widetilde{B_2}(T).$$

To combine these estimates, we set

$$A(T) := 2\max\left(A_1(T), A_2, A_3(T), A_4(T)\right) \qquad \text{and} \qquad B(T) := 2\max\left(B_1(T), B_2(T)\right)$$

and sum them to get

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\|u\|^2_{L^2(\Gamma)} + \|u_b\|^2_{L^2(\Gamma)}\right) \leq A(T)\left(\|u\|^2_{L^2(\Gamma)} + \|u_b\|^2_{L^2(\Gamma)}\right) + B(T)\,\|v\|^2_{H^1(\Omega)}$$

for almost all t. An application of Gronwall's lemma gives

$$
\begin{aligned}
\|u(t)\|^2_{L^2(\Gamma)} + \|u_b(t)\|^2_{L^2(\Gamma)} &\leq e^{tA(T)}\left[\|u(0)\|^2_{L^2(\Gamma)} + \|u_b(0)\|^2_{L^2(\Gamma)} + B(T)\int_0^t \|v(s)\|^2_{H^1(\Omega)}\,\mathrm{d}s\right] \\
&\leq B(T)e^{TA(T)}\,\|v\|^2_{L^2(0,\widetilde{T};H^1(\Omega))}\,,
\end{aligned}
$$

since the initial conditions of $\overline{u}$ and $\widetilde{u}$ are the same, and therefore $u(0) = 0$ for every $x \in \Gamma$. Naturally, the same holds for $u_b(0)$. After integrating from 0 to $\widetilde{T}$, we can conclude that

$$\|u\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \|u_b\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} \leq \widetilde{T}B(T)e^{TA(T)}\,\|v\|^2_{L^2(0,\widetilde{T};H^1(\Omega))}\,.$$

It may seem like we are done now, but the current estimate uses the wrong norm on v. To get an estimate using the $L^2(0,\widetilde{T};L^2(\Omega))$-norm instead, we can use Equation (2.11) to get

$$
\begin{aligned}
&\|u\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \|u_b\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} \\
&\leq \widetilde{T}B(T)e^{TA(T)}D_\varepsilon(T)\left(\|u\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \|u_b\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))}\right) + \overbrace{\widetilde{T}B(T)e^{TA(T)}\,\|v\|^2_{L^2(0,\widetilde{T};L^2(\Omega))}}^{\geq 0},
\end{aligned}
\tag{2.12}
$$

or after rearranging terms,

$$\|u\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} + \|u_b\|^2_{L^2(0,\widetilde{T};L^2(\Gamma))} \leq \frac{\widetilde{T}B(T)e^{TA(T)}}{1 - \widetilde{T}B(T)e^{TA(T)}D_\varepsilon(T)}\,\|v\|^2_{L^2(0,\widetilde{T};L^2(\Omega))}\,.$$

As this factor clearly approaches 0 when $\widetilde{T}$ goes to 0, we can choose $\widetilde{T}$ sufficiently small such that it is less than 1 and conclude the proof. □

We have now gathered the needed auxiliary results to prove the main statement of this section.

Proof of Theorem 2.2. Let $\widetilde{T} > 0$ be small enough such that, by Lemmas 2.5 and 2.6, $\mathcal{T}_1$ and $\mathcal{T}_2$ are contractions with Lipschitz constants q_1 and q_2, respectively. Define, as discussed before, $\mathcal{T}$ as the composition of $\mathcal{T}_1$ and $\mathcal{T}_2$. Then for any $\overline{v}, \widetilde{v} \in L^2(0,\widetilde{T}; L^2(\Omega))$, we have

$$\|\mathcal{T}(\overline{v}) - \mathcal{T}(\widetilde{v})\|^2_{L^2(0,\widetilde{T};L^2(\Omega))} \leq q_2 \|\mathcal{T}_1(\overline{v}) - \mathcal{T}_1(\widetilde{v})\|^2_{\left(L^2(0,\widetilde{T};L^2(\Gamma))\right)^2} \leq q_2 q_1 \|\overline{v} - \widetilde{v}\|^2_{L^2(0,\widetilde{T};L^2(\Omega))},$$

showing that $\mathcal{T}$ is a contraction since $q_2 q_1 \in [0,1)$. We apply Banach's fixed-point theorem to get a unique fixed point $v \in L^2(0,\widetilde{T}; L^2(\Omega))$. As previously discussed, this corresponds to a unique weak solution on $(0,\widetilde{T})$.

We now seek to extend this to a solution on $(0,T)$. Since all bounds for u, u_b and v were derived independently of $\widetilde{T}$, we can set new initial conditions

$$\widetilde{v}_0 := v(\tfrac{1}{2}\widetilde{T}) \qquad \text{and} \qquad (\widetilde{u}_0, \widetilde{u}_{b,0}) = (u(\tfrac{1}{2}\widetilde{T}), u_b(\tfrac{1}{2}\widetilde{T})) := \mathcal{T}_1(v)(\tfrac{1}{2}\widetilde{T})$$

and get the same contraction estimates. This gives another local solution $\widetilde{v}$, this time defined on $(\frac{1}{2}\widetilde{T}, \frac{3}{2}\widetilde{T})$. Because of the uniqueness properties of both v and $\widetilde{v}$, their overlap on $(\frac{1}{2}\widetilde{T}, \widetilde{T})$ must coincide. This means that $\widetilde{v}$ can be used to define an extension of v to the interval $(0, \frac{3}{2}\widetilde{T})$.

This process can be repeated, getting a solution on $(\frac{j}{2}\widetilde{T}, \frac{j+2}{2}\widetilde{T})$ for $j = 2, \ldots$ and using them to define extensions of v until we cover the full time interval $(0,T)$. We can simultaneously construct the corresponding solutions to the ODEs by extending the local solutions using $(\widetilde{u}, \widetilde{u}_b) := \mathcal{T}_1(\widetilde{v})$ at every step. For these solutions u, u_b and v, the same non-negativity and boundedness properties hold as for all the local solutions, since these properties were proven independently of $\widetilde{T}$. $\square$

Chapter 3

Turing patterns

Morphogenesis is the biological process in the development of organisms that is concerned with organising the cells and tissue into shapes and patterns. In [Tur52], Alan Turing discussed a class of reaction-diffusion systems modelling chemical substances reacting with each other and diffusing through a tissue, and suggested that these dynamics could be sufficient to explain the main mechanisms of morphogenesis. The development of these patterns, referred to as *Turing patterns* [Mur03], has since then been used in a wide range of applications in biology [Mur03, EOP+12] as well as other fields of science [ZPVB21]. We want to investigate whether this phenomenon can be used to account for the heterogeneous distribution of receptors on the cell membrane. In this chapter, we will briefly discuss the analytical theory of this pattern formation process and provide two examples of pattern forming systems that are relevant for our models discussed in Chapters 2 and 4.

3.1 Stability analysis

We give an outline of the derivation of necessary conditions for the existence of Turing patterns given in [Mur03]. Consider the concentrations of two chemical substances. Their dynamics are modelled by two coupled reaction-diffusion equations. Let $\Omega \subseteq \mathbb{R}^n$ be an open and bounded set with C^1 boundary and $T > 0$. Let $u(t,x)$ and $v(t,x)$ denote the concentrations of the two species at $t \in [0,T]$ and $x \in \overline{\Omega}$. A reaction-diffusion model for these concentrations u and v is of the form

$$\begin{aligned}
u_t &= \Delta u + \gamma f(u,v) && \text{in } (0,T)\times\Omega,\\
v_t &= d\Delta v + \gamma g(u,v) && \text{in } (0,T)\times\Omega,\\
\frac{\partial u}{\partial \nu} &= \frac{\partial v}{\partial \nu} = 0 && \text{on } [0,T)\times\partial\Omega,\\
u(0,\cdot\,) &= u_0 && \text{in } \Omega,\\
v(0,\cdot\,) &= v_0 && \text{in } \Omega,
\end{aligned} \tag{3.1}$$

where $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}$ are the reaction functions, $d > 0$ is the diffusion coefficient for v, $\gamma > 0$ is a reaction scaling coefficient and $u_0, v_0 : \Omega \to \mathbb{R}$ are given initial data. Turing patterns can emerge when this type of system has a homogeneous equilibrium that is linearly stable in the absence of diffusion, i.e. if it is a linearly stable steady state of the system

$$u_t = \gamma f(u, v), \qquad v_t = \gamma g(u, v), \tag{3.2}$$

and becomes unstable due to the introduction of diffusion in Equation (3.1), referred to as a *diffusion driven instability*. The relevant steady state is the positive solution (U, V) of

$$f(U, V) = g(U, V) = 0,$$

if it exists. To first investigate the stability of the steady state in absence of diffusion, we linearise the system of Equation (3.2) around (U, V) and compute the eigenvalues of the corresponding matrix. For the steady state to be asymptotically stable, we require that $\operatorname{Re}\lambda < 0$ for every eigenvalue $\lambda \in \mathbb{C}$ of this matrix, for which we should have

$$f_u + g_v < 0 \qquad \text{and} \qquad f_u g_v - f_v g_u > 0, \tag{3.3}$$

where all partial derivatives are evaluated at the steady state (U, V). To study the instability of the steady state with respect to the system of Equation (3.1), the system is linearised and decomposed into *Fourier modes* w_k with *wave number* k[1]. We assume that the solution can be written as

$$\begin{pmatrix} u(t, x) - U \\ v(t, x) - V \end{pmatrix} = w(t, x) = \sum_k c_k e^{\lambda(k)t} w_k(x),$$

for coefficients $c_k \in \mathbb{R}$ and growth rates $\lambda(k) \in \mathbb{C}$ for every wave number k. If the growth rate $\lambda(k)$ has positive real part for some wave number k, the corresponding Fourier mode grows exponentially in time, causing the steady state to become unstable. Which Fourier modes precisely cause the instability depends on γ and influences the spatial scale of the patterns.

One can show that these considerations lead to the following additional set of necessary conditions for the reaction functions f and g,

$$df_u + g_v > 0 \qquad \text{and} \qquad (df_u + g_v)^2 - 4d\,(f_u g_v - f_v g_u) > 0, \tag{3.4}$$

where all partial derivatives are evaluated at (U, V). In particular, the conditions of Equations (3.3) and (3.4) require f_u and g_v to be of opposite sign, and $d > 1$. Chemically, this is interpreted as one of the two species diffusing faster than the other. Lastly, the reaction scaling coefficient γ should be set such that the spatial scale of the patterns is comparable to the size of the spatial domain Ω.

3.2 Example: the Schnakenberg equations

The Schnakenberg equations, originally discussed in [Sch79], form an example of a pattern forming reaction-diffusion system that is relevant for our models in Chapters 2 and 4. Their reaction functions are given by

$$f(u, v) = a - u + u^2 v \qquad \text{and} \qquad g(u, v) = b - u^2 v, \tag{3.5}$$

[1]These Fourier nodes and wave numbers are the solutions of the problem $\Delta w_k + k^2 w_k = 0$. They depend on the dimension n, the shape of the spatial domain Ω as well as the boundary conditions of Equation (3.1). For more details, see [Mur03].

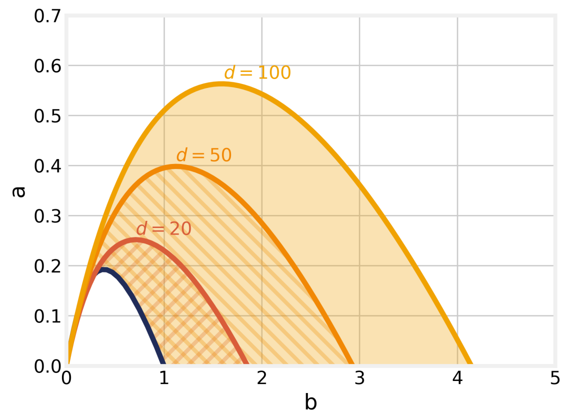

Figure 3.1: The Turing space for Equation (3.5) plotted for $d = 20$, $d = 50$ and $d = 100$.

where a and b are positive real numbers representing constant production of both chemical species. In [Mur03], a detailed analysis is given into the specific conditions for these reactions to form Turing patterns in one-dimensional domains. As a result, we get conditions for the parameters in the model that define a subset of the parameter space within which Turing instabilities take place, referred to as the *Turing space*. For fixed d, this subset is given by the area between the two curves

$$a > \frac{1}{2}U\left(1 - U^2\right), \qquad b = \frac{1}{2}U\left(1 + U^2\right),$$
$$a < \frac{1}{2}U\left(1 - \frac{2U}{\sqrt{d}} - \frac{U^2}{d}\right), \qquad b = \frac{1}{2}U\left(1 + \frac{2U}{\sqrt{d}} + \frac{U^2}{d}\right),$$

parameterised by U, see Figure 3.1. If (a, b) lies between these two curves and γ is high enough, a Turing instability will take place. You can see that the Turing space increases if larger values of d are chosen. In Figures 3.2 and 3.3, a simulation of this model is shown with parameters within the Turing space. In this simulation, the domain is $\Omega = (-8, 8)^2$ and the parameters are set to $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The initial conditions are set to

$$u_0(x) = U + F(x) \qquad \text{and} \qquad v_0(x) = V + G(x), \tag{3.6}$$

where

$$U := a + b \qquad \text{and} \qquad V := \frac{b}{(a+b)^2} \tag{3.7}$$

are the corresponding uniform steady states and F and G are random perturbations whose values are uniformly sampled from $(-0.1, 0.1)$. For more details on this perturbation of the uniform steady states, see Chapter 4.

3.3 Example: Schnakenberg with decay

The reaction function f in Equation (3.5) includes natural decay of the species u. As it is relevant for the models we consider in Chapters 2 and 4, we show that the addition of decay of v in the

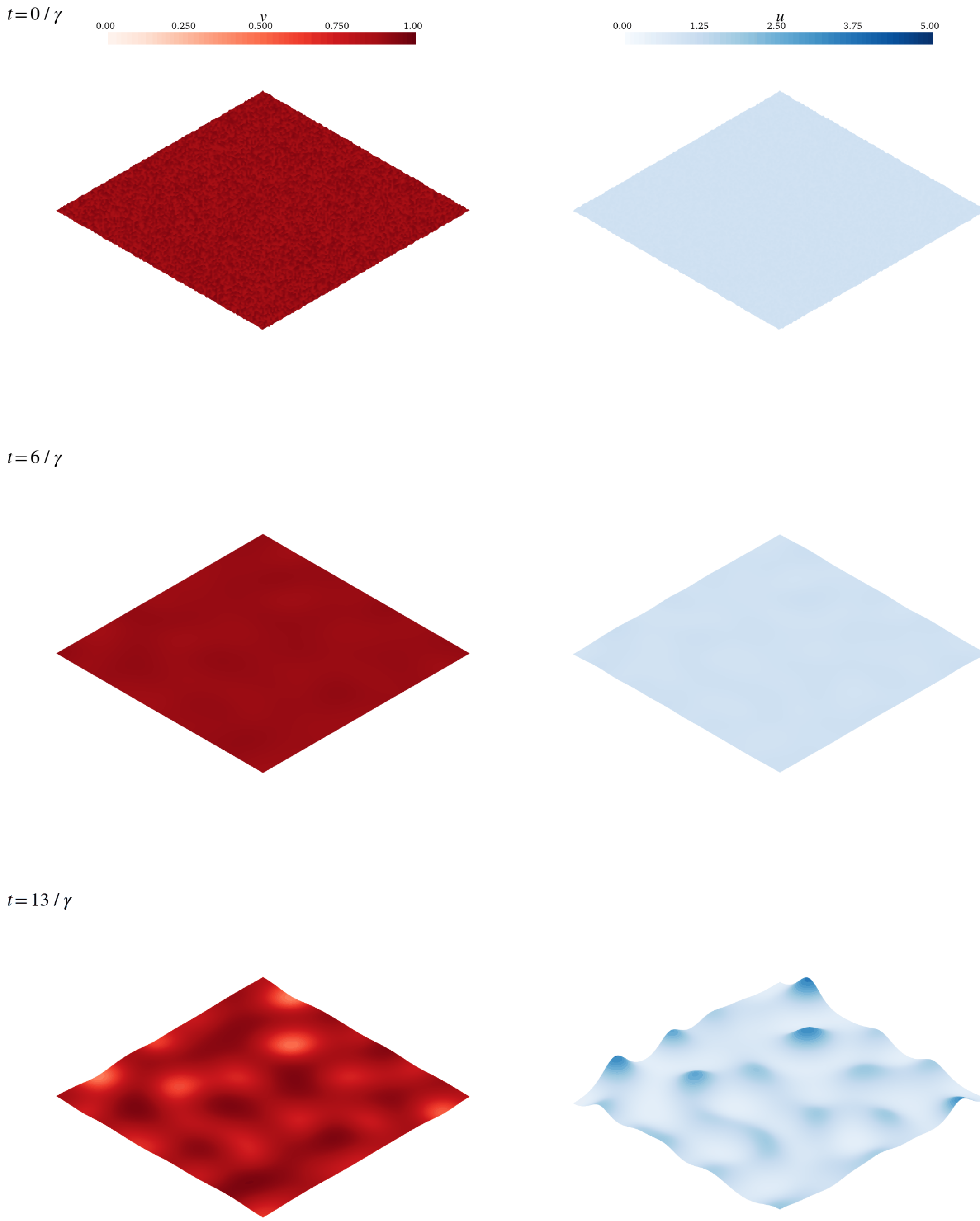

Figure 3.2: A simulation of the Schnakenberg equations on $\Omega = (-8,8)^2$ with $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is plotted for times $t = 0$, $t = 5/\gamma$ and $t = 11/\gamma$.

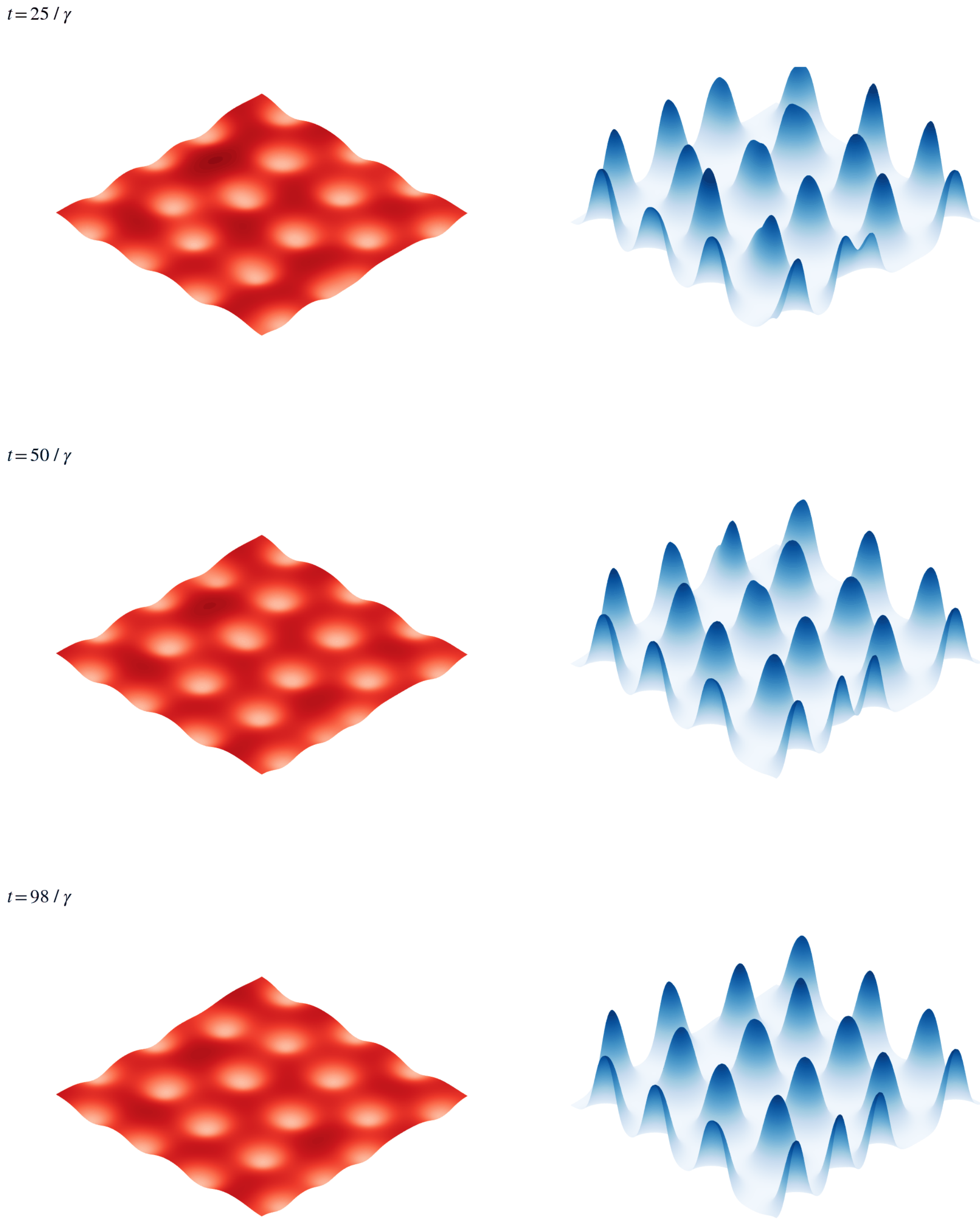

Figure 3.3: The same simulation of the Schnakenberg equations on $\Omega = (-8, 8)^2$ with $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is now plotted for times $t = 14$, $t = 20/\gamma$ and $t = 80/\gamma$.

function g does not cause the model to lose its ability to generate Turing patterns. To this end, we consider the kinetics

$$f(u,v) = a - u + u^2 v \qquad \text{and} \qquad g(u,v) = b - \delta_v v - u^2 v, \tag{3.8}$$

where $\delta_v > 0$ is the decay rate for v. The new uniform steady states are U and V with U being a positive real root of

$$U^3 - (a+b)U^2 + \delta_v U - a\delta_v = 0,$$

if such a root exists, and

$$V := \frac{b}{U^2 + \delta_v}.$$

One can check that if $\delta_v = 0$, we recover the uniform steady states from the classical Schnakenberg equations, given in Equation (3.7). When we use these new kinetics in the model of Equation (3.1) with the new parameter δ_v set to 0.1, we see from the simulations in Figures 3.3 and 3.4 that this model is also able to generate Turing patterns. In this simulation, the other parameters are set to the same values as before, i.e. $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The domain is $\Omega = (-8,8)^2$ again, and the initial conditions are set to a random perturbation around the uniform steady states as in Equation (3.6). With this relatively low value of δ_v, we see that this model generates patterns that are similar to the ones generated by the classical Schnakenberg equations in both shape and scale.

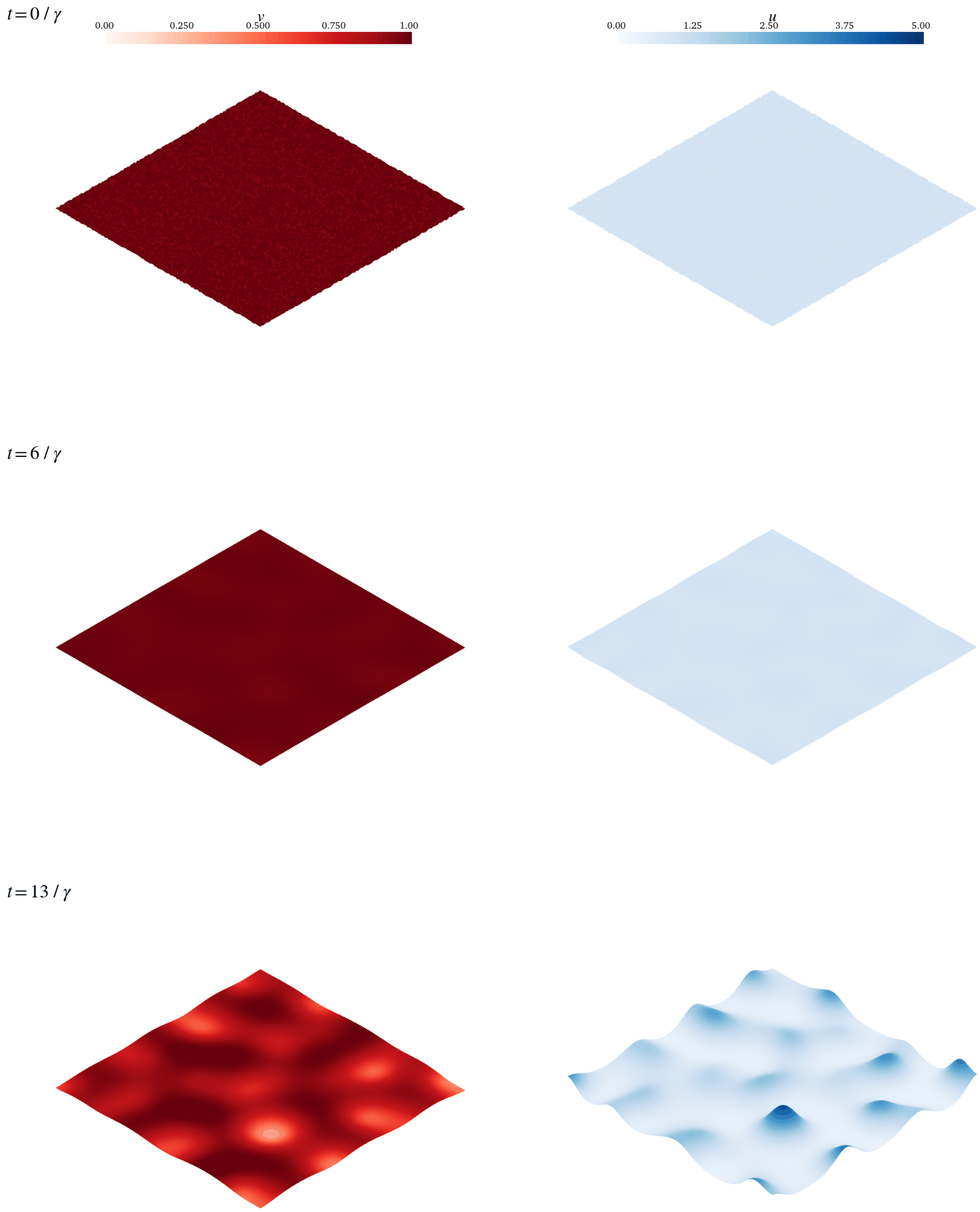

Figure 3.4: A simulation of the Schnakenberg equations with extra decay for v on $\Omega = (-8,8)^2$ with $\delta_v = 0.1$, $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is plotted for times $t = 0$, $t = 5/\gamma$ and $t = 11/\gamma$.

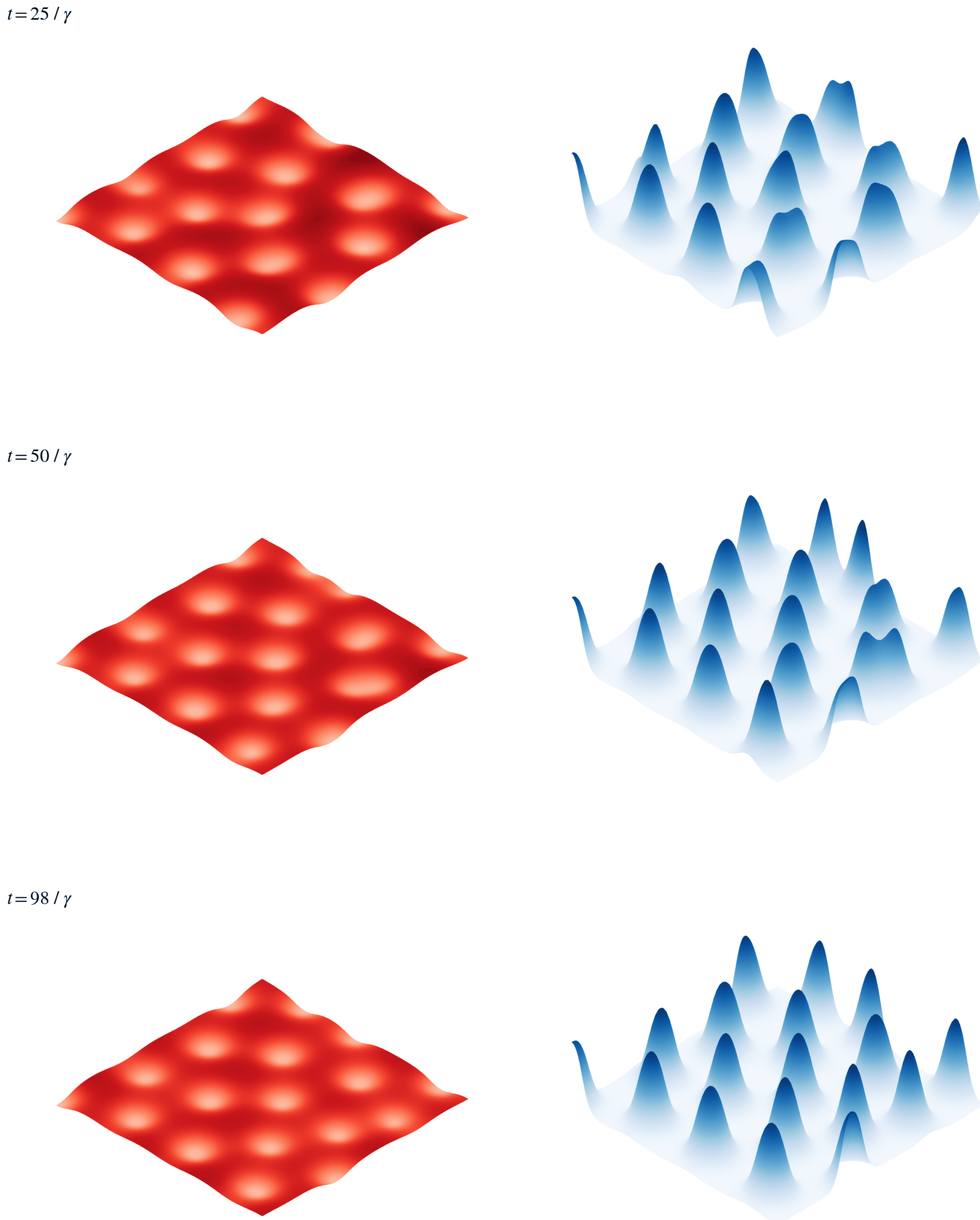

Figure 3.5: The same simulation of the Schnakenberg equations with extra decay for v on $\Omega = (-8,8)^2$ with $\delta_v = 0.1$, $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is now plotted for times $t = 16$, $t = 20/\gamma$ and $t = 80/\gamma$.

Chapter 4

Numerical analysis

We have proven the existence and uniqueness and some qualitative properties of solutions to our model. Some properties, however, are more difficult to investigate using purely analytical methods. One of these properties is the ability to generate Turing patterns. We therefore turn to numerical methods and simulate the solutions of the model to have a more concrete and clear view of its behaviour and study if the models can predict receptor clustering on the cell membrane.

4.1 Modified model

4.1.1 Diffusion on the surface

Now that we are working with simulations, we can make our model slightly more complex by dropping one of our assumptions. Let Ω and Γ be as in Section 2.1. In this chapter, we restrict our study to the dimensions of $d = 2$ and $d = 3$, since these are the most relevant for the biological applications of this model. Recall that in Section 2.1, we assumed the receptors, both free and bounded, to be fixed in a location on the cell membrane. This resulted in a system of one PDE and two ODEs, the latter being both defined at every point on the cell membrane. We now drop this assumption and add diffusion of free and bounded receptors, leading to the following system of reaction diffusion PDEs

$$
\begin{aligned}
u_t &= D_u \Delta u + \gamma\left(P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 v + 2k_{\text{off}} u_b\right) && \text{on } (0,T)\times\Gamma,\\
u_{b,t} &= D_{u_b} \Delta u_b + \kappa\gamma\left(-\delta_{u_b} u_b + k_{\text{on}} u^2 v - k_{\text{off}} u_b\right) && \text{on } (0,T)\times\Gamma,\\
v_t &= D_v \Delta v - \gamma\delta_v v && \text{in } (0,T)\times\Omega,\\
D_v \frac{\partial v}{\partial \nu} &= \gamma\left(P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b\right) && \text{on } (0,T)\times\Gamma,\\
D_v \frac{\partial v}{\partial \nu} &= 0 && \text{on } (0,T)\times\Gamma_e,
\end{aligned}
\tag{4.1}
$$

where D_u and D_{u_b} are positive. We have also added two reaction scaling coefficients $\gamma > 0$ and $\kappa > 0$. When Turing instabilities can take place in the system, we will adjust γ such that the domain is sufficiently large and the patterns become visible. The second scaling coefficient κ is used to simulate the quasi-steady state assumption we will discuss in the next subsection. The corresponding initial conditions are

$$u(0,x) = u_0(x), \qquad u_b(0,x) = u_{b,0}(x) \qquad \text{and} \qquad v(0,y) = v_0(y)$$

for all $x \in \Gamma$ and $y \in \overline{\Omega}$. These initial conditions are non-negative.

There are multiple things to note about System (4.1). Since Γ has an empty interior as a subset of $\mathbb{R}^d$, it is impossible to define spatial derivatives for functions exclusively defined on this set. This means we would not be able to use the Laplace operator to include diffusion of the receptors as we do in System (4.1). This issue is solved by using the so called the Laplace-Beltrami operator instead, which is a generalisation of the Laplace operator in $\mathbb{R}^d$ to Riemannian manifolds. Since we assumed Γ to be a C^1 boundary, it forms such a manifold. The Laplace-Beltrami operator is further discussed in [Jos17], but we will not go into details here. For our purposes, it is sufficient to note that it behaves similarly to the classical Laplace operator and admits analogous rules of calculus such as partial integration. It is sometimes denoted by Δ_Γ, but for the sake of simplicity, we will omit this notation and use Δ for both the Laplace and Laplace-Beltrami operators, but they are in fact different operators.

Since we now have PDEs for u and u_b, it seems necessary to impose certain boundary conditions. However, recalling our exact construction of the domain, we can see that the set Γ representing the cell membranes, when viewed as a separate $(d-1)$-dimensional manifold, does not have a boundary. Since the equations for u and u_b strictly live on this manifold, boundary conditions are not required.

4.1.2 Derivation of the two-species model

We seek to compare this three-species bulk-surface model to a two-species model similar to that derived in [KMI14]. To reduce our model to a two-species model, we make some of the assumptions as in [KMI14]. The first and most impactful is that the dynamics of the bounded receptors are so fast that $u_{b,t}$ and Δu_b are negligible. In the simulations, this will be realised by setting $\kappa \gg 1$. This results in an explicit expression for u_b in terms of u and v,

$$u_b = \frac{k_{\text{on}}}{\delta_{u_b} + k_{\text{off}}} u^2 v.$$

This is called a *quasi-steady state*. Using this expression for u_b enables us to simplify System (4.1) to the two-species model

$$\begin{aligned}
u_t &= D_u \Delta u + \gamma\left(P_u - \delta_u u + \beta u^2 v\right) && \text{on } (0,T)\times\Gamma,\\
v_t &= D_v \Delta v - \gamma \delta_v v && \text{in } (0,T)\times\Omega,\\
D_v \frac{\partial v}{\partial \nu} &= \gamma\left(P_v - \beta u^2 v\right) && \text{on } (0,T)\times\Gamma,\\
D_v \frac{\partial v}{\partial \nu} &= 0 && \text{on } (0,T)\times\Gamma_e,
\end{aligned} \tag{4.2}$$

where $\beta := \frac{k_{\text{on}}\delta_{u_b}}{\delta_{u_b}+k_{\text{off}}}$ and we make the second assumption, namely that $\alpha = 3\beta$. The final assumption made in [KMI14] is that $\delta_v v \ll \beta u^2 v$, making the natural decay of the ligands negligible and recovering the classical Schnakenberg equations. However, as we saw in Section 3.3, the Schnakenberg equations with additional decay of v are also able to generate patterns, so we will not be making this last assumption.

4.2 The Finite Element Method

Numerical methods for ODEs typically include defining a small time step to discretise the equation and using the ODE to derive a suitable iteration. In the study of PDEs, one might think to extend this approach to the spatial variable. This would result in defining a time step $\tau > 0$ as well as a space discretisation $h > 0$ and discretising the domain $(0, T) \times \Omega$ as a subset of $\tau\mathbb{Z} \times h\mathbb{Z}^d$. One method that uses this approach is the *finite difference* method, which is very successfully applied to a wide range of PDEs. However, this method is fairly hard to use when the dimension gets higher than $d = 1$ or $d = 2$, or the shape of the domain becomes more complex than, for instance, a rectangle. Our model is defined on a domain that contains holes that are designed to model cells and as a result, will contain some curvature. The finite difference method is too unreliable in this case [QSS07], so we need a more robust and adaptable approach.

4.2.1 Basic construction

The *finite element* method is another way to approximate solutions to a PDE. It is mainly used for discretising the space variable, so let us focus on an example that does not include time. We consider the standard Poisson equation

$$\Delta u = f \qquad \text{in } \Omega \tag{4.3}$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \qquad \text{on } \partial\Omega. \tag{4.4}$$

Here, $\Omega \subseteq \mathbb{R}^d$ is an open and bounded set with C^1 boundary and f and g are known real-valued functions on Ω and $\partial\Omega$, respectively. We first derive the weak form of this PDE. Multiply Equation (4.3) with a smooth function $\varphi \in C^\infty(\overline{\Omega})$ and integrate to get

$$\int_\Omega f\varphi = \int_\Omega \Delta u \varphi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu}\varphi \, \mathrm{d}S - \int_\Omega \nabla u \cdot \nabla\varphi = \int_{\partial\Omega} g\varphi \, \mathrm{d}S - \int_\Omega \nabla u \cdot \nabla\varphi.$$

A weak solution of Equations (4.3) and (4.4) then is a function $u \in H^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla\varphi = \int_{\partial\Omega} g\varphi \, \mathrm{d}S - \int_\Omega f\varphi \tag{4.5}$$

for every $\varphi \in H^1(\Omega)$.

By discretising the spatial domain, we will transform Equation (4.5) into a system of linear equations. To this end, we use the same approach as in [DE13] and define a *mesh*. We choose a finite number of points $\mathcal{N}_h = \{x_i\}_{i=1}^m$ in $\overline{\Omega}$ and use them as vertices to define d-simplices[1], which we collect in a set $\mathcal{T}_h$. These points are called *nodes*, and the d-simplices they enclose are referred to as *elements*. To have a well-behaved discretisation, for any two distinct d-simplices $T, T' \in \mathcal{T}_h$ we require $T \cap T'$ to either be empty or form a common subsimplex of dimension $k < d$.

[1] An n-simplex is the n-dimensional equivalent of a triangle. The most important instances are the 0-simplex, which is a point, the 1-simplex, a line segment, the 2-simplex, a triangle, and the 3-simplex, a tetrahedron. Every n-simplex is enclosed by $n+1$ $(n-1)$-simplices, which in turn are enclosed by $(n-2)$-simplices, etc. All of these are referred to as its subsimplices.

This leaves us with an approximation of Ω, denoted by

$$\Omega_h := \bigcup_{T \in \mathcal{T}_h} T.$$

The subscript h derives from the idea that this still represents the fineness of our discretisation, just as in the finite difference method. If we define

$$h := \max_{T \in \mathcal{T}_h} \operatorname{diam}(T),$$

we want Ω_h to approximate Ω increasingly well as h approaches 0.

Additionally, we want a discretised function space that we use to construct approximate solutions to Equations (4.3) and (4.4), called the *trial space*. To this end, for every node $x_i \in \mathcal{N}_h$, we define a basis function $\varphi_i : \Omega_h \to \mathbb{R}$ such that $\varphi_i(x_j) = \delta_{ij}$ for all $j \in \{1, \dots, m\}$ and φ_i is linearly interpolated on each element. Now the function space we consider for our trial functions is the space of linear combinations of these basis functions,

$$\mathcal{V}_{\Omega_h} := \operatorname{span}\{\varphi_1, \dots, \varphi_m\},$$

which is a finite-dimensional subspace of $H^1(\Omega)$. This is called the space of *first-order Lagrange elements*, derived from the fact that the functions are piecewise linear polynomials on each element. There are many other kinds of function spaces used in finite element simulations, such as second-order Lagrange spaces, where the functions are piecewise quadratic polynomials on each element, or spaces where the functions are allowed to be discontinuous, referred to as discontinuous Galerkin elements [CJST98].

We can now use this setup to convert our original problem into one that a computer can handle. Let $u_h \in \mathcal{V}_{\Omega_h}$ be an approximate solution to Equation (4.5). That is, let u_h solve

$$\int_{\Omega_h} \nabla u_h \cdot \nabla \varphi = \int_{\partial\Omega_h} g\varphi \,\mathrm{d}S - \int_{\Omega_h} f\varphi$$

for every $\varphi \in \mathcal{V}_{\Omega_h}$[2]. Then for any $i \in \{1, \dots, m\}$, we can set $\varphi = \varphi_i$ and write u_h as its linear decomposition with coefficients $U_1, \dots, U_m \in \mathbb{R}$, leading to

$$\int_{\partial\Omega_h} g\varphi_i \,\mathrm{d}S - \int_{\Omega_h} f\varphi_i = \int_{\Omega_h} \nabla \left(\sum_{j=1}^{m} U_j \varphi_j \right) \cdot \nabla\varphi_i = \sum_{j=1}^{m} U_j \int_{\Omega_h} \nabla\varphi_j \cdot \nabla\varphi_i.$$

If we define

$$F_i := \int_{\partial\Omega_h} g\varphi_i \,\mathrm{d}S - \int_{\Omega_h} f\varphi_i \qquad \text{and} \qquad K_{ij} := \int_{\Omega_h} \nabla\varphi_j \cdot \nabla\varphi_i$$

for every $i, j \in \{1, \dots, m\}$, this is equivalent to the linear system

$$KU = F. \tag{4.6}$$

Here, K is typically referred to as the *stiffness matrix*. We have now transformed the PDE into a linear system of equations, which we can solve using a wide variety of algorithms.

[2]Note that for this to be well-defined, we should also define some approximation of g and f such that they are properly defined on $\partial\Omega_h$ and Ω_h, respectively. We will not go into detail about that here.

4.2.2 Time-dependent PDEs

We should also illustrate how time discretisation ties into this approach. Consider, therefore, the heat equation

$$u_t = \Delta u + f \qquad \text{in } (0,T) \times \Omega \tag{4.7}$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \qquad \text{on } [0,T) \times \partial\Omega, \tag{4.8}$$

where $T > 0$ is some final time and now f and g are real-valued functions on $(0,T) \times \Omega$ and $[0,T) \times \partial\Omega$, respectively. Of course, we also need initial conditions

$$u(0,x) = u_0(x), \qquad x \in \Omega. \tag{4.9}$$

From Section 1.5, we know that we can define a weak solution of this PDE as a function $u \in H^1(0,T;H^1(\Omega))$ such that Equation (1.6) holds for every $\varphi \in H^1(\Omega)$ and almost every $t \in [0,T]$.

Similarly to the numerical study of ODEs, we now choose a number of time steps $N \in \mathbb{N}$ and let $\tau := T/N$. For every $n \in \{1,\dots,N\}$, we seek to approximate the solution at the time step $n\tau$,

$$u_n(x) := u(n\tau, x), \qquad x \in \Omega.$$

To this end, we use a difference quotient to approximate the time derivative and write

$$\int_\Omega \frac{u_{n+1} - u_n}{\tau}\varphi + \int_\Omega \nabla u_{n+1} \cdot \nabla\varphi = \int_{\partial\Omega} g(n\tau,x)\varphi(x)\,\mathrm{d}S(x) + \int_\Omega f(n\tau,x)\varphi(x)\,\mathrm{d}x.$$

where we have used an implicit scheme for diffusion and an explicit one for the forcing. This is a widely used scheme, commonly referred to as IMEX [ARS97, PR01, BPR25]. Other schemes can be used with varying results in numerical stability and accuracy, depending on what class of PDE it is applied to. In this thesis, we will not go further into details about this choice.

After rearranging terms, this leads to the time-independent problem

$$\int_\Omega u_{n+1}\varphi + \tau\int_\Omega \nabla u_{n+1} \cdot \nabla\varphi = \tau\int_\Gamma g(n\tau,x)\varphi(x)\,\mathrm{d}S(x) + \int_\Omega u_n\varphi + \tau\int_\Omega f(n\tau,x)\varphi(x)\,\mathrm{d}x \tag{4.10}$$

for $n = 0,\dots,N-1$. After substituting u_{n+1} and u_n with their linear decompositions and setting $\varphi = \varphi_i \in \mathcal{V}_{\Omega_h}$ for some $i \in \{1,\dots,m\}$, this is transformed to the system of equations

$$(M + \tau K)U^{n+1} = MU^n + \tau F^n,$$

where $U^n \in \mathbb{R}^m$ contains the coefficients for the linear decomposition of u_n into $\varphi_1,\dots,\varphi_m$, and F^n and M are defined by

$$F_i^n := \int_{\partial\Omega_h} g(n\tau,x)\varphi_i(x)\,\mathrm{d}S(x) + \int_{\Omega_h} f(n\tau,x)\varphi_i(x)\,\mathrm{d}x \qquad \text{and} \qquad M_{ij} := \int_{\Omega_h} \varphi_j\varphi_i$$

for $i,j \in \{1,\dots,m\}$. The matrix M is often called the *mass matrix*. When calculating a new iteration U^{n+1}, we can assume the previous iterations are known, starting off with the initial conditions

$$U_i^0 := u_0(x_i), \qquad i = 0,\dots,m.$$

In this example, we first discretised time, leading to a PDE in space which was then discretised with the finite element method. This order can also be swapped by assuming that the approximate solution u can be written as a linear combination of functions in $\mathcal{V}_{\Omega_h}$ with time-dependent coefficients

$$u(t,x) = \sum_{j=1}^{m} U_j(t)\varphi_j(x).$$

Then the process of integrating over Ω and using integration by parts leads to a system of linear ODEs for $U(t) = (U_1(t), \dots, U_m(t))$, called the *semi-discrete* form of the PDE [Tho84]. This approach is more similar to the Galerkin method for proving existence and uniqueness of weak solutions to PDEs as demonstrated in [Eva10]. The reason why we instead choose to illustrate discretising time first is that, as we will discuss later, the libraries we use to simulate the PDEs require us to input the form of Equation (4.10), after which it will derive the linear system of equations for us. We therefore first discretise time instead of first discretising space.

4.2.3 Bulk-surface equations

In a bulk-surface setting, the interacting functions are defined on different spatial domains of different dimensions, so we should carefully think about how to discretise them. Let $\Gamma \subseteq \partial\Omega$. When simulating a bulk-surface equation, we first discretise the bulk and extract the surface mesh from the bulk mesh by defining the set of *boundary facets* (or *exterior facets*)

$$\mathcal{F}_h^e := \{S \mid S \text{ is a } (d-1)\text{-dimensional subsimplex of some } T \in \mathcal{T}_h \text{ and every vertex of } S \text{ lies on } \Gamma\},$$

after which we define

$$\Gamma_h := \bigcup_{S\in\mathcal{F}_h^e} S \subseteq \partial\Omega_h,$$

analogous to the definition of Ω_h. This also generates a subset of boundary nodes $\mathcal{N}_h^\Gamma := \mathcal{N}_h \cap \Gamma_h = \{y_1, \dots, y_l\}$, on which we can define another collection of basis functions $\psi_1, \dots, \psi_l$ such that $\psi_i(y_j) = \delta_{ij}$ for every $i, j \in \{1, \dots, l\}$ and every ψ_i is linearly interpolated in between nodes. This defines a finite-dimensional subspace of $H^1(\Gamma)$ given by

$$\mathcal{V}_{\Gamma_h} := \operatorname{span}\{\psi_1, \dots \psi_l\}.$$

In the next section, this function space is used to derive the time discretisations for our bulk surface models.

4.3 Implementation

We now derive the time discretisations of Systems (4.2) and (4.1). For both models, we use an IMEX scheme where we choose implicit discretisation for all linear terms and explicit discretisation for the non-linear reaction. For System (4.2), this results in the iteration

$$(1+\tau\gamma\delta_u)\int_{\Gamma_h} u_{n+1}\psi \,\mathrm{d}S + \tau D_u \int_{\Gamma_h} \nabla u_{n+1}\cdot\nabla\psi \,\mathrm{d}S = \int_{\Gamma_h} u_n\psi \,\mathrm{d}S + \tau\gamma \int_{\Gamma_h} \left[P_u + \beta u_n^2 v_n\right]\psi \,\mathrm{d}S \quad (4.11)$$

for the surface PDE, where we now let Γ_h denote the submesh of $\partial\Omega_h$ that approximates Γ, and

$$(1+\tau\gamma\delta_v)\int_{\Omega_h} v_{n+1}\varphi + \tau D_v \int_{\Omega_h} \nabla v_{n+1}\cdot\nabla\varphi = \int_{\Omega_h} v_n\varphi\,\mathrm{d}S - \tau\gamma\int_{\Gamma_h}\left[P_u + \beta u_n^2 v_n\right]\varphi\,\mathrm{d}S \tag{4.12}$$

for the bulk PDE, where $\psi \in \mathcal{V}_{\Gamma_h}$ and $\varphi \in \mathcal{V}_{\Omega_h}$. With the same approach, System (4.1) admits the time discretisation

$$\begin{aligned}(1+\tau\gamma\delta_u)\int_{\Gamma_h} u_{n+1}\psi\,\mathrm{d}S + \tau D_u\int_{\Gamma_h}\nabla u_{n+1}\psi\,\mathrm{d}S - \tau\gamma(\alpha+2k_{\text{off}})\int_{\Gamma_h} u_{b,n+1}\psi\,\mathrm{d}S \\ = \int_{\Gamma_h} u_n\psi\,\mathrm{d}S + \tau\gamma\int_{\Gamma_h}\left[P_u - 2k_{\text{on}}u_n^2 v_n\right]\psi\,\mathrm{d}S\end{aligned} \tag{4.13}$$

for the PDE for u,

$$\begin{aligned}(1+\tau\kappa\gamma\delta_{u_b})\int_{\Gamma_h} u_{b,n+1}\chi\,\mathrm{d}S + \tau D_{u_b}\int_{\Gamma_h}\nabla u_{b,n+1}\chi\,\mathrm{d}S + \tau\kappa\gamma k_{\text{off}}\int_{\Gamma_h} u_{b,n+1}\chi\,\mathrm{d}S \\ = \int_{\Gamma_h} u_{b,n}\chi\,\mathrm{d}S + \tau\kappa\gamma k_{\text{on}}\int_{\Gamma_h} u_n^2 v_n\chi\,\mathrm{d}S\end{aligned} \tag{4.14}$$

for the PDE for u_b, and finally for v,

$$\begin{aligned}(1+\tau\gamma\delta_v)\int_{\Omega_h} v_{n+1}\varphi + \tau D_v\int_{\Omega_h}\nabla v_{n+1}\varphi - \tau\gamma k_{\text{off}}\int_{\Gamma_h} u_{b,n+1}\varphi\,\mathrm{d}S \\ = \int_{\Omega_h} v_n\varphi + \tau\gamma\int_{\Gamma_h}\left[P_v - k_{\text{on}}u_n^2 v_n\right]\varphi\,\mathrm{d}S,\end{aligned} \tag{4.15}$$

where $\psi,\chi \in \mathcal{V}_{\Gamma_h}$ and $\varphi \in \mathcal{V}_{\Omega_h}$.

The initial conditions are set as random perturbations of the uniform steady states. That is, we define $u_0, u_{b,0} \in \mathcal{V}_{\Gamma_h}$ and $v_0 \in \mathcal{V}_{\Omega_h}$ by

$$u_0 = \sum_{i=1}^{l}(U+a_i)\psi_i, \qquad u_{b,0} = \sum_{i=1}^{l}(U_b+b_i)\psi_i \qquad \text{and} \qquad v_0 = \sum_{i=1}^{m}(V+c_i)\varphi_i,$$

where U, U_b and V are the uniform steady states for u, u_b and v, respectively, and $a_1,\dots,a_l$, $b_1,\dots,b_l$ and $c_1,\dots,c_m$ are uniformly sampled from $[-0.1, 0.1]$.

To calculate these iterations, Equations (4.11) to (4.15) are given as input to Python scripts using FEniCSx, along with a mesh generated by Gmsh. FEniCSx derives the system of linear equations, which is solved with PETSc.

4.3.1 Code availability

All scripts that were written and used for this thesis can be found on the public GitHub page https://github.com/bregtvdl03/Bachelor_Thesis_Turing_patterns.

4.4 Results

We first simulate solutions to Systems (4.2) and (4.1) for $d = 2$ with Ω being the open square $(-32, 32)^2$ without the four closed disks of radius 4 and centers $(\pm 8, \pm 8)$ and Γ being the boundaries of these disks. We further set $D_v = 50$, $D_u = 1$, $\delta_v = 0.05$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $\beta = 1$ and $\gamma = 64$. In Figures 4.1 and 4.2, the numerical solution of System (4.2) is plotted at different time steps. We can clearly see Turing patterns appearing on the cell membranes, similar to the ones generated by the Schnakenberg equations in Sections 3.2 and 3.3. These now represent clusters of receptors, with the regions of lower densities representing relative scarcity.

We compare this to the numerical solutions of System (4.1). To simulate the quasi-steady state assumption, we set $\kappa = 50$ and choose parameters D_{u_b}, δ_{u_b}, k_{on} and k_{off} such that $\beta = \frac{k_{\text{on}}\delta_{u_b}}{\delta_{u_b}+k_{\text{off}}} = 1$. For this example, we chooose $D_{u_b} = 1$, $\delta_{u_b} = 1$, $k_{\text{on}} = 20$ and $k_{\text{off}} = 19$. The additional reactions scaling coefficient κ necessitates increasing the number of steps N significantly to preserve numerical stability. If the model of two species proves to be a good representation of the three-species model, it would be preferable to use this model rather than the full three-species equivalent, since it is simpler to analyse both numerically and analytically. In Figures 4.3 and 4.4, the numerical solution of System (4.1) is plotted for similar time steps. It admits the same general behaviour as the numerical solution of System (4.2), smoothing out the perturbations before forming Turing patterns of similar scale and shape. The patterns formed by u and u_b almost coincide, with the peaks of u_b being a slightly higher than those of u. This can be justified by the extra scaling coefficient κ dominating the diffusion.

For $d = 3$, the domain is defined as the open cube $(-32, 32)^3$ without the eight closed spheres of radius 4 and centers $(\pm 8, \pm 8, \pm 8)$, and Γ is defined as the boundary of these spheres. We use the same parameters as the previous simulations, except δ_v is set to 0.01. In Figures 4.5 and 4.6, we see the two-species system form patterns on the cell membranes and their effect on the concentration of ligands in the immediate neighbourhood of the cell membranes. Again, this behaviour is very similar to the numerical solution of the three-species model, shown in Figures 4.7 and 4.8. The peaks of u are also similar to that of u_b again, with the peaks of u_b being slightly higher than those of u. Note that in the simulation of Figures 4.7 and 4.8, the initial conditions are set to the uniform steady state with larger perturbations. However, this does not significantly affect the behaviour of the pattern formation.

For both $d = 2$ and $d = 3$ and the chosen parameter values, the model of System (4.2) seems to be a good representation for that of System (4.1). Moreover, when choosing the two-species model to further study the behaviour of these PDEs, the concentration of u_b can be assumed to be similar to that of u. These conclusions only hold for the chosen parameter values. To extend them to the general case, more numerical tests should be done, since the parameters in System (4.2) leave three degrees of freedom for choosing the parameters in System (4.1). Hence, every choice of parameters in System (4.2) admits an infinite set of parameters for System (4.1) such that the quasi-steady state reduction produces the same model.

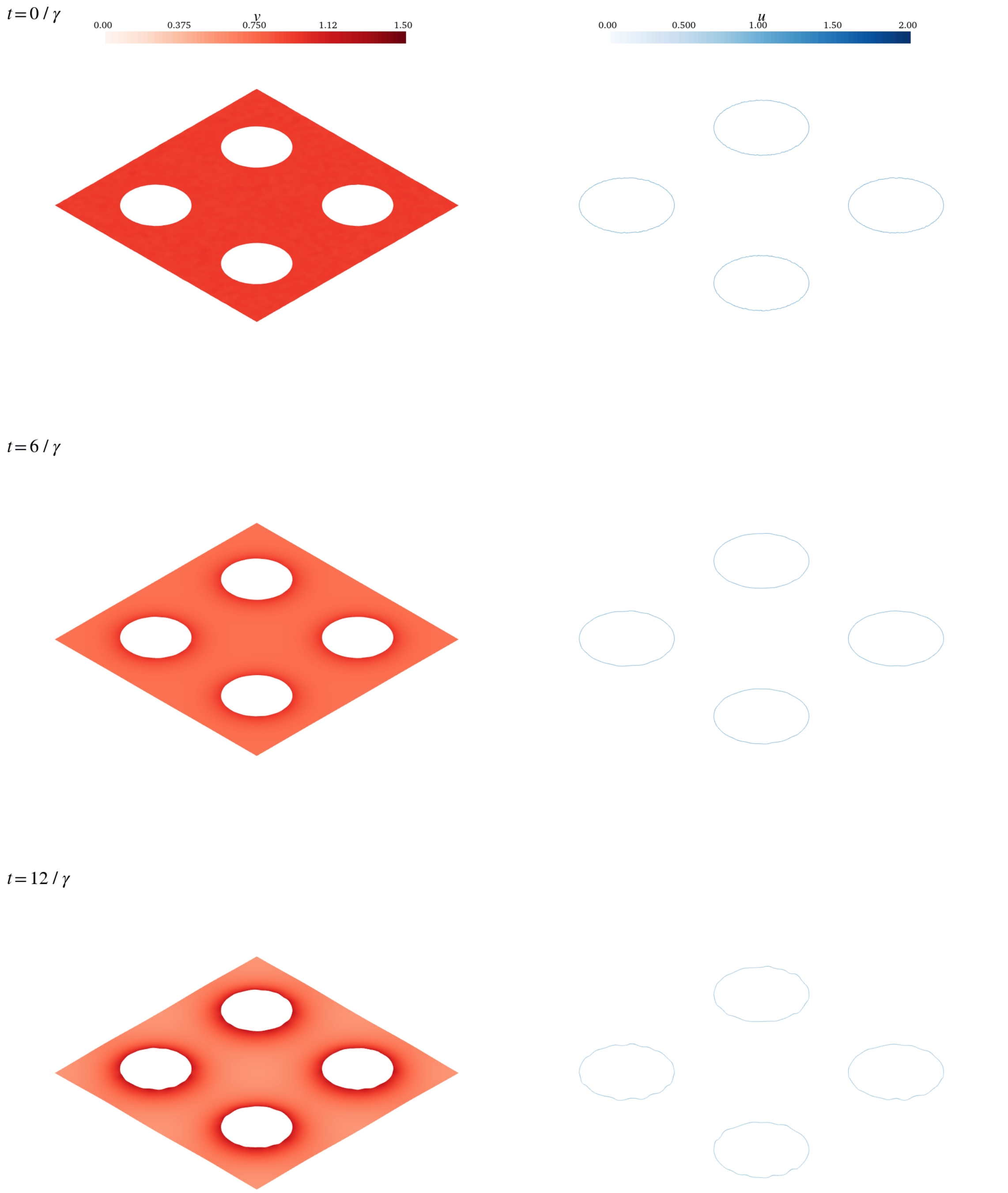

Figure 4.1: A simulation of System (4.2) on $(-32, 32)^2$ with four disks with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.05$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $\beta = 1$ and $\gamma = 64$. The solution is plotted for times $t = 0$, $t = 6/\gamma$ and $t = 12/\gamma$.

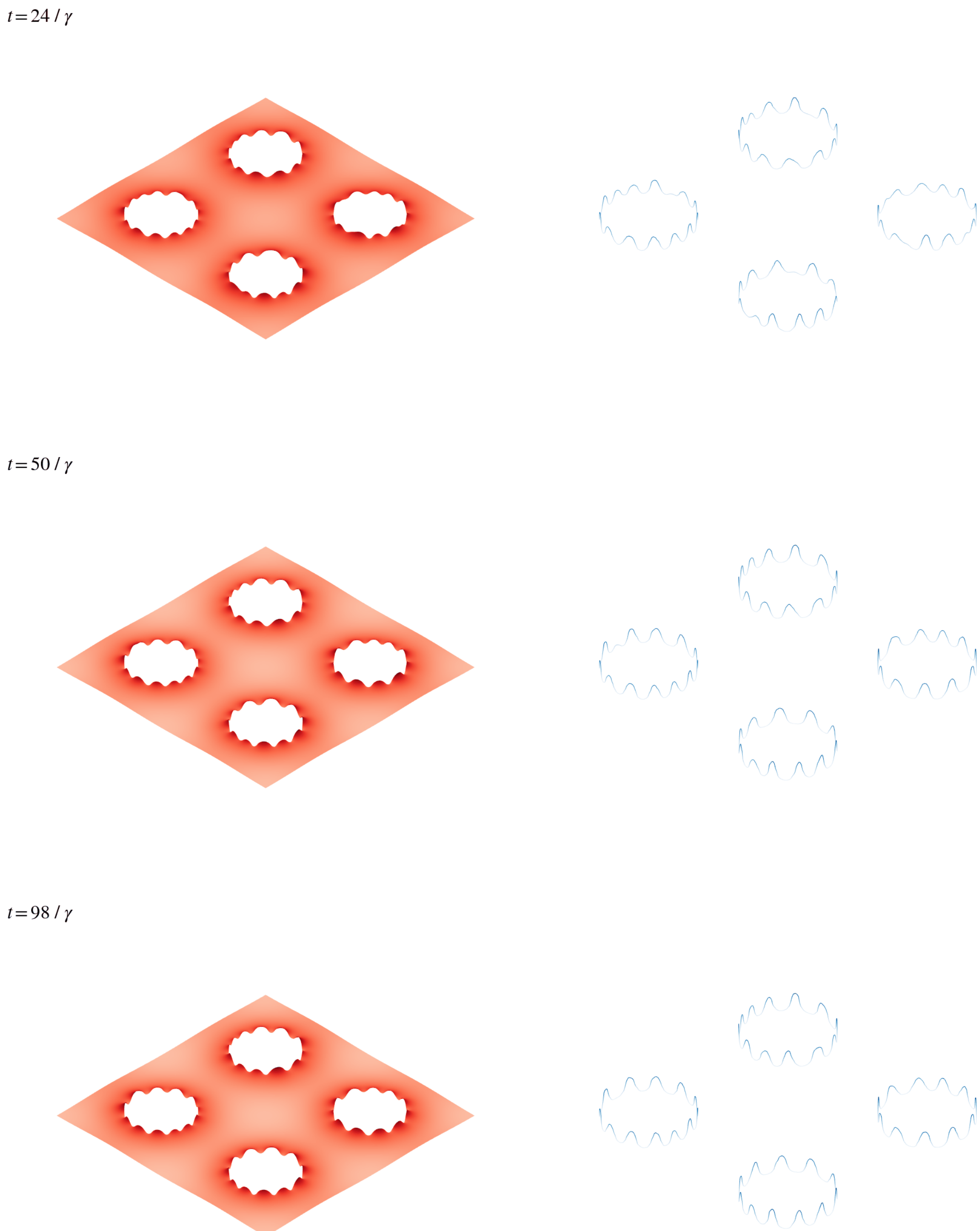

Figure 4.2: A simulation of System (4.2) on $(-32, 32)^2$ with four disks with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.05$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $\beta = 1$ and $\gamma = 64$. The solution is plotted for times $t = 24/\gamma$, $t = 50/\gamma$ and $t = 98/\gamma$.

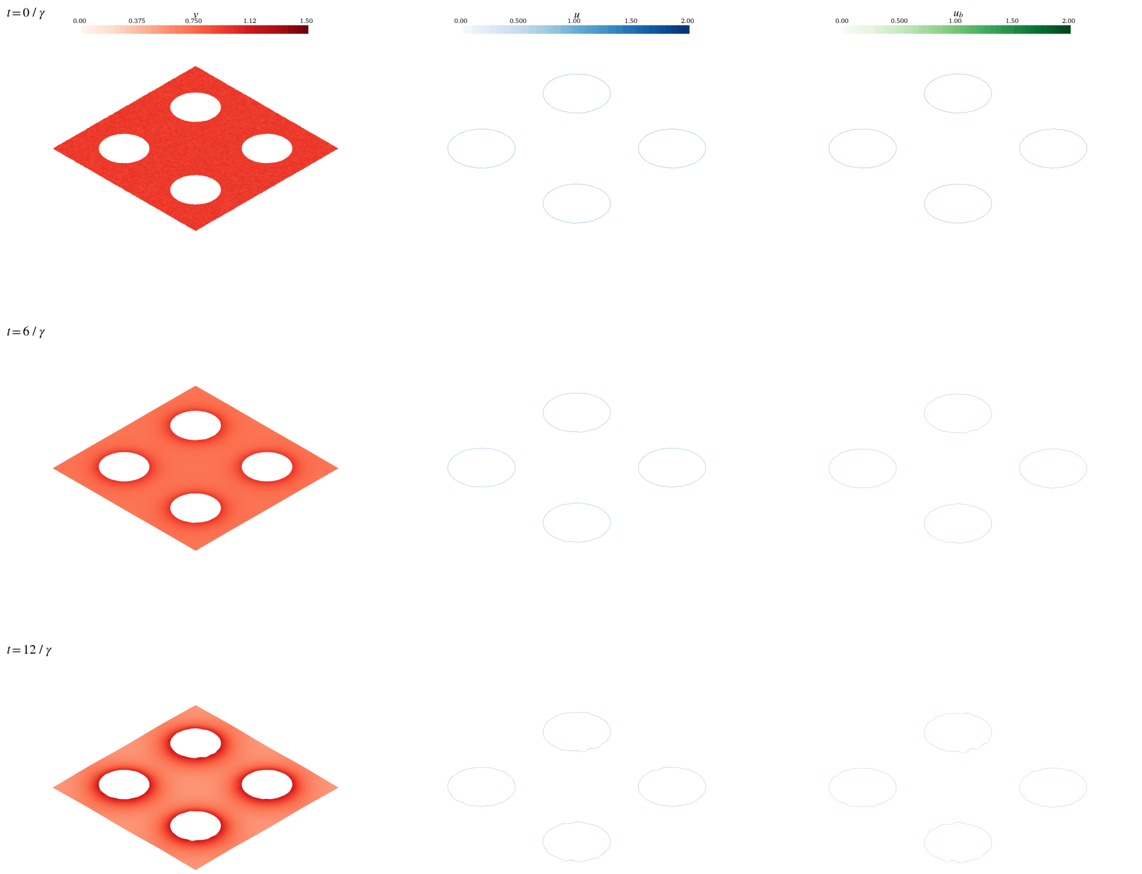

Figure 4.3: A simulation of System (4.1) on $(-32, 32)^2$ with four disks with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.05$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $D_{u_b} = 1$, $\delta_{u_b} = 1$, $k_{\text{on}} = 20$ and $k_{\text{off}} = 19$, $\kappa = 50$ and $\gamma = 64$. The solution is plotted for times $t = 0$, $t = 6/\gamma$ and $t = 12/\gamma$.

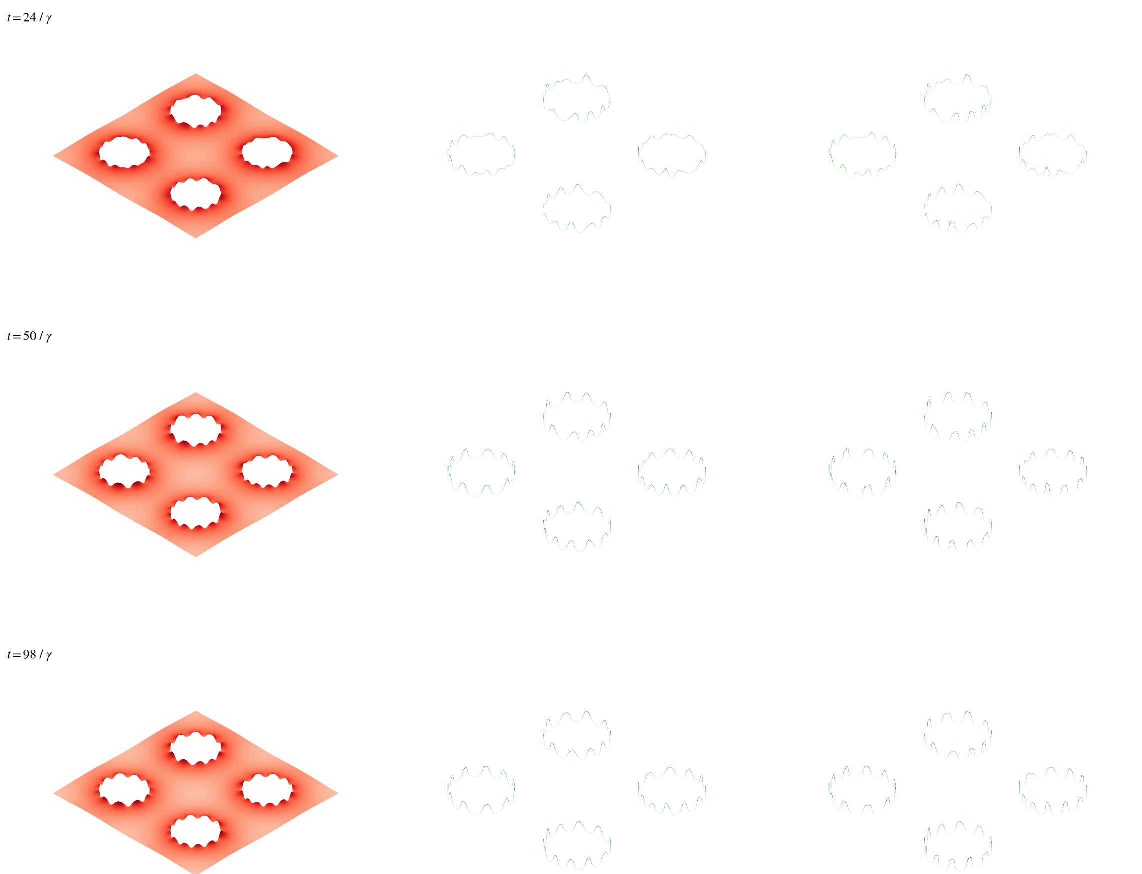

Figure 4.4: A simulation of System (4.1) on $(-32, 32)^2$ with four disks with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.05$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $D_{u_b} = 1$, $\delta_{u_b} = 1$, $k_{\text{on}} = 20$ and $k_{\text{off}} = 19$, $\kappa = 50$ and $\gamma = 64$. The solution is plotted for times $t = 24/\gamma$, $t = 50/\gamma$ and $t = 98/\gamma$.

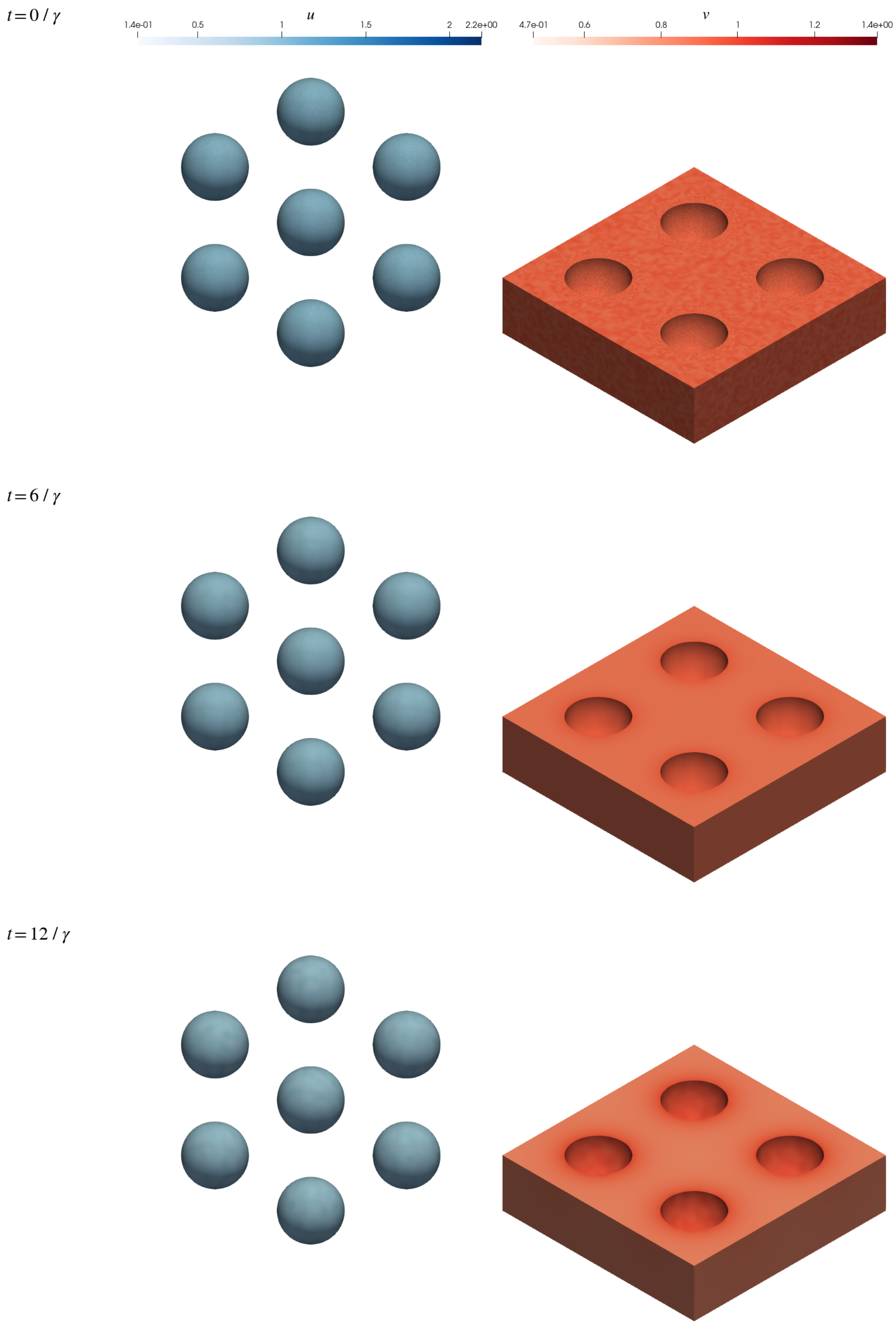

Figure 4.5: A simulation of System (4.2) on $(-32, 32)^3$ with eight spheres with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.01$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $\beta = 1$ and $\gamma = 64$. The solution is plotted for times $t = 0$, $t = 6/\gamma$ and $t = 12/\gamma$.

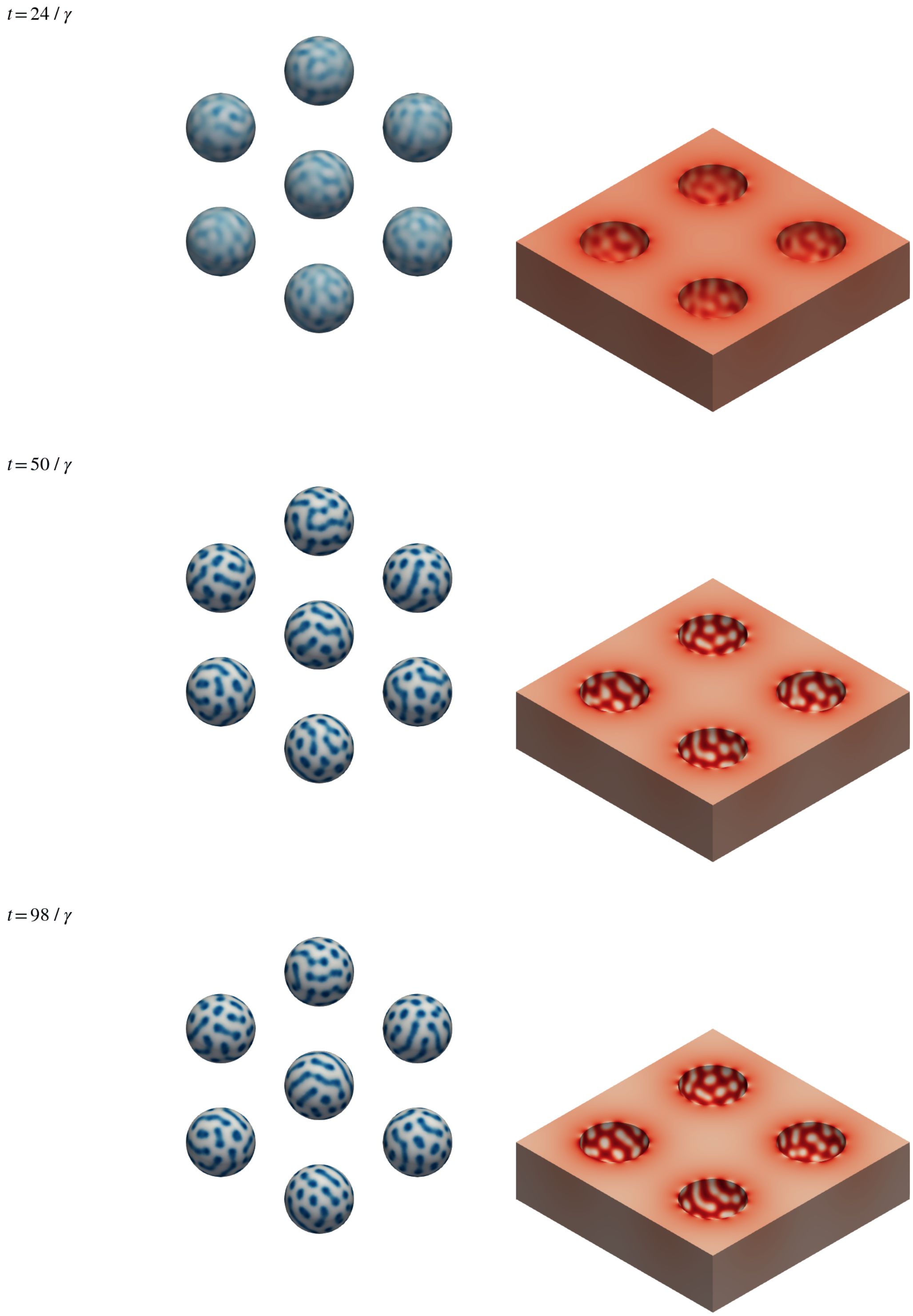

Figure 4.6: A simulation of System (4.2) on $(-32, 32)^3$ with eight spheres with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.01$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $\beta = 1$ and $\gamma = 64$. The solution is plotted for times $t = 24/\gamma$, $t = 50/\gamma$ and $t = 98/\gamma$.

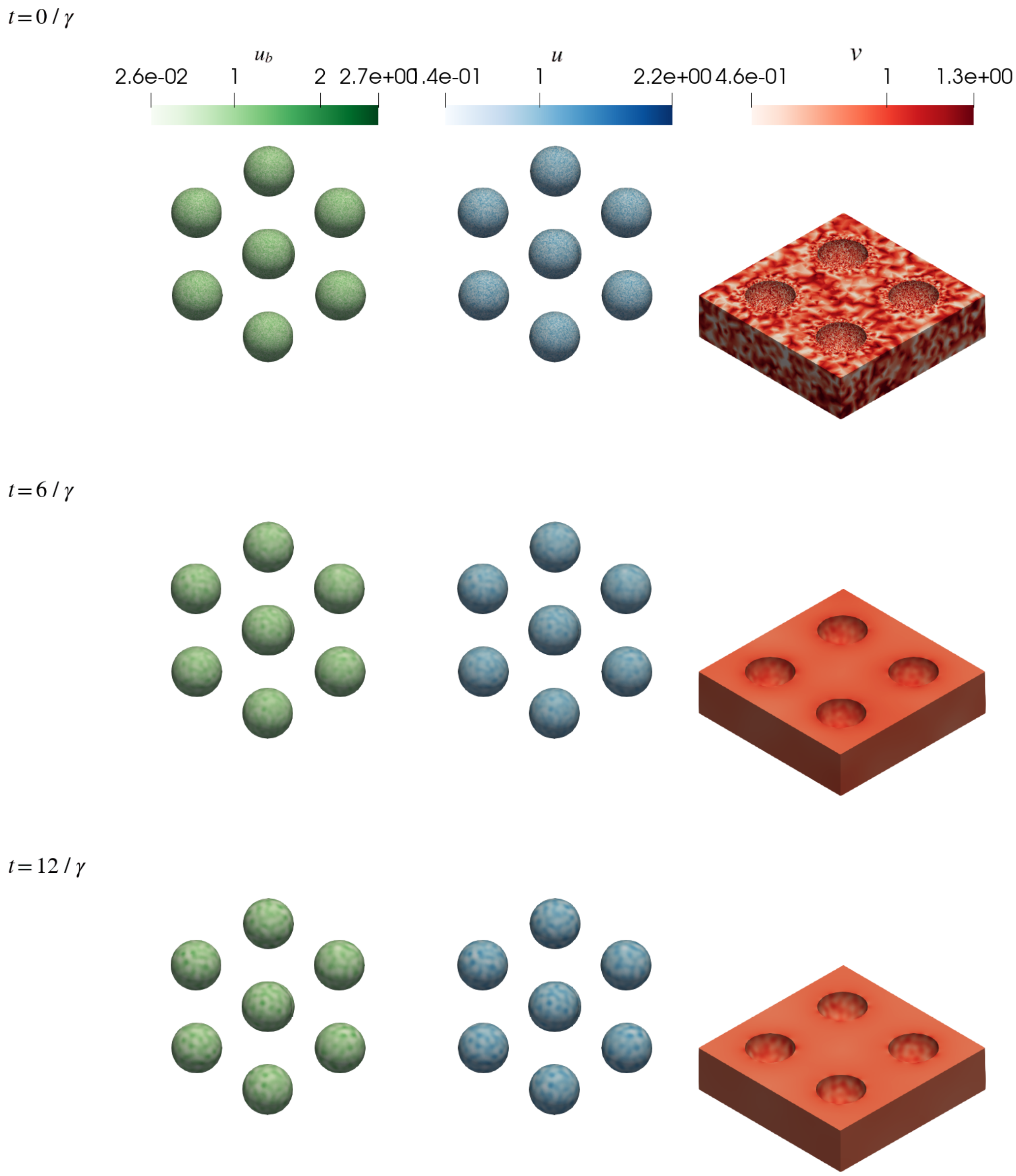

Figure 4.7: A simulation of System (4.1) on $(-32, 32)^3$ with eight spheres with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.01$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $D_{u_b} = 1$, $\delta_{u_b} = 1$, $k_{\text{on}} = 20$ and $k_{\text{off}} = 19$, $\kappa = 50$ and $\gamma = 64$. The solution is plotted for times $t = 0$, $t = 6/\gamma$ and $t = 12/\gamma$.

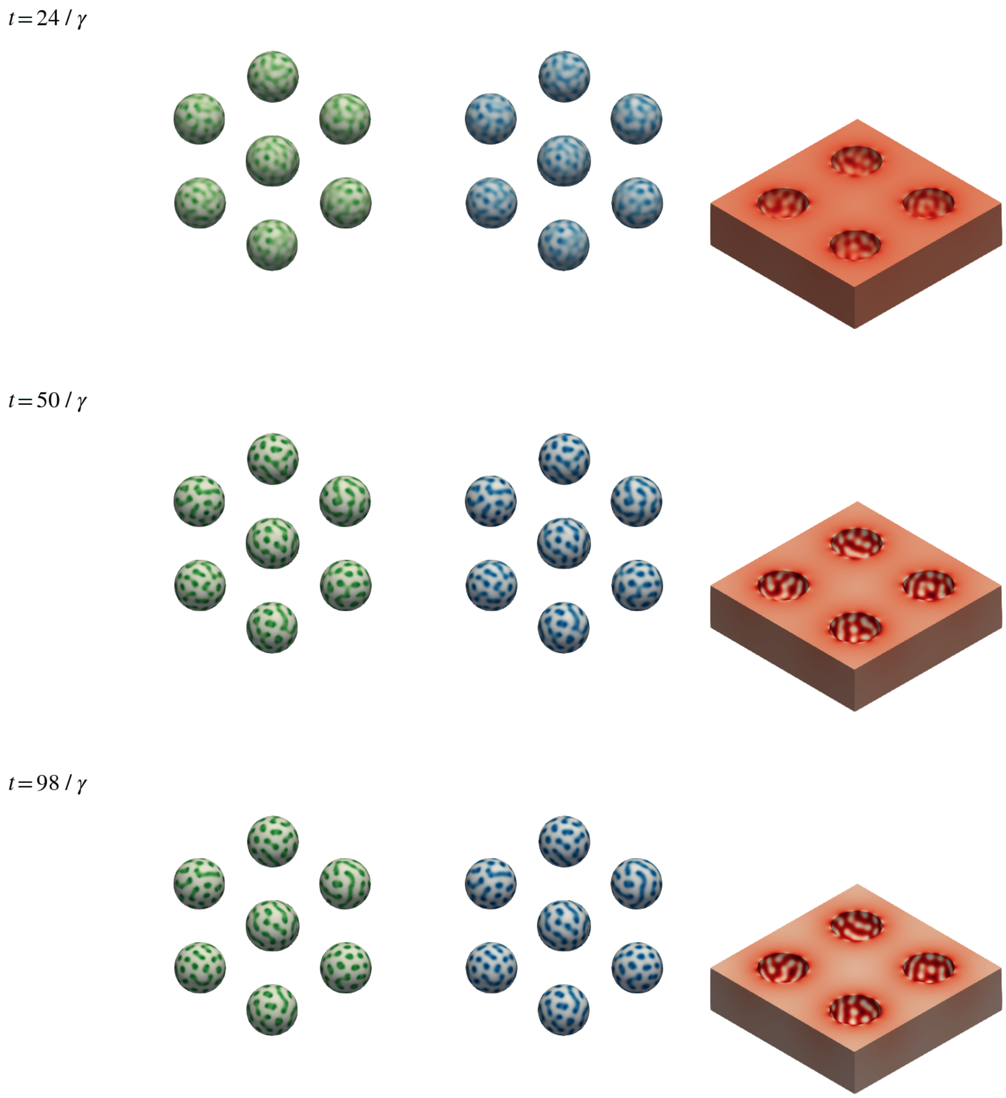

Figure 4.8: A simulation of System (4.1) on $(-32, 32)^3$ with eight spheres with radius 4 cut out and parameters set to $D_v = 50$, $D_u = 1$, $\delta_v = 0.01$, $\delta_u = 1$, $P_u = 0.1$, $P_v = 0.9$, $D_{u_b} = 1$, $\delta_{u_b} = 1$, $k_{\text{on}} = 20$ and $k_{\text{off}} = 19$, $\kappa = 50$ and $\gamma = 64$. The solution is plotted for times $t = 24/\gamma$, $t = 50/\gamma$ and $t = 98/\gamma$.

Conclusion and outlook

In this thesis, we have discussed different bulk-surface systems modelling interactions of ligands and receptors. Using a fixed-point argument, we have seen that weak solutions to the ODE-PDE system of Chapter 2 exist for every finite time interval and that they are unique. Furthermore, we have proven that these solutions are bounded and non-negative on these finite time intervals, which are both mathematically desirable and biologically meaningful.

We then studied how spatial organisations can arise in ligand-receptor systems through the theory of Turing patterns. This provided a theoretical explanation for how receptors can be found distributed in clusters despite their diffusive nature. Building on this theory, we introduced a modified model that incorporates diffusion on the cell membrane and derived a reduced two-species formulation that is simpler to analyse and simulate.

Finally, we discussed the finite element method and used it to numerically solve two different bulk-surface reaction-diffusion equations in two and three dimensional domains. These simulations showed that the modified model is able to generate Turing patterns. We reduced the model to a two-species variant and showed that this reduced form reproduces the behaviour of the full system while requiring less computational effort. This makes the two-species system an adequate alternative for future analytical and numerical studies.

There are multiple interesting directions for further research. One could drop the assumption of Chapter 2 that free and bounded receptors are fixed in their position on the cell membrane and instead study the existence and uniqueness of solutions of System (4.1). Additionally, the regularity of these solutions could be studied, as well as their long-time behaviour such as convergence towards a steady state. It could also be interesting to derive a constant bound for these solutions for infinite time intervals, inspired by the observations in Chapter 4. To ensure the accuracy of conclusions drawn from the numerical solutions, one could study stability and convergence properties of the finite element method applied to these models. Finally, comparing the predictions of these models with experimental data would help further develop them as useful tools for biological studies.

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